

Algorithmic Pricing and Competition: Empirical Evidence from the German Retail Gasoline Market

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We provide the first empirical analysis of the relationship between algorithmic pricing (AP) and competition by studying the impact of adoption in Germany's retail gasoline market, where software became widely available in 2017. Because adoption dates are unknown, we identify adopting stations by testing for structural breaks in AP markers, finding most breaks to be around the time of widespread AP introduction. Because station adoption is endogenous, we instrument using headquarter adoption. Adoption increases margins but only for nonmonopoly stations. In duopoly and triopoly markets, margins increase only if all stations adopt, suggesting that AP has a significant effect on competition.

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I. Introduction

Pricing algorithm technology has become increasingly sophisticated in recent years. Although firms have made use of pricing software for decades, technological advancements have created a shift from mechanically set prices to AI-powered algorithms that can handle large quantities of data and interact, learn, and make decisions with unprecedented speed and sophistication. The evolution of algorithmic pricing (AP) software has raised concerns regarding possible impact on firm behavior and competition. The potential for algorithms to facilitate collusion, either tacit or explicit, has been a popular discussion point among antitrust authorities, economic organizations, and competition law experts in recent years (Ezrachi and Stucke 2015, 2016, 2017; OECD 2017; Competition Bureau 2018; Varian 2018; Agrawal, Gans, and Goldfarb 2019; Autorité de la Concurrence and Bundeskartellamt 2019; UK Digital Competition Expert Panel 2019). Since the goal of algorithms is to converge to an optimal policy, AI agents could learn to play a collusive strategy to achieve a joint profit-maximizing outcome. AP software can also facilitate collusion through increased ease of monitoring and speed of detection and through punishment of possible deviations.

The literature on algorithmic collusion is expanding, with contributions from economics, law, and computer science. At present, there is no theoretical consensus as to whether algorithms facilitate tacit collusion (Kühn and Tadelis 2018; Miklós-Thal and Tucker 2019; Calvano et al. 2020; Asker, Fershtman, and Pakes 2021; Brown and MacKay 2023). Despite some evidence that collusive algorithmic behavior can appear in synthetic environments, there are questions about whether it can and will arise in practice. As of yet, there is no empirical evidence linking the adoption and use of pricing algorithms to market outcomes related to competition. The objective of this paper is to supplement existing theoretical literature by

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conducting the first empirical analysis of the impact of wide-scale adoption of AP software. We focus on the German retail gasoline market, where, according to trade publications and news articles, AP software became widely available beginning in 2017 and for which we have access to a high-frequency database of prices and characteristics for every retail gas station in the country from January 2016 to December 2018.¹

Investigating the impact of the adoption of algorithmic pricing software on competition requires overcoming three important challenges. First, even with access to detailed pricing data, adoption decisions are typically not publicly observed. Second, adoption is endogenous, since the decision to adopt is correlated with factors that are unobserved to the researcher. Finally, even if adoption can be causally linked with higher prices or margins, it is not clear whether these can be attributed to changes in competition intensity rather than to other factors, such as an improved ability to detect underlying fluctuations in wholesale prices or predict demand.

To overcome the first challenge, we test for structural breaks in pricing behaviors that are thought to capture the promised impacts of sophisticated pricing software: (1) the number of price changes made in a day, (2) the response time of a station's price to a rival's price change, (3) the responsiveness of a station's price to crude oil shocks, and (4) the responsiveness of a station's price to local weather shocks. We focus on these measures since leading providers characterize their software as performing high-frequency analysis to "rapidly, continuously, and intelligently" react to market conditions. For each measure, we test for structural breaks at each station, considering each week in a large window around the time of supposed adoption (Quandt 1960). For each measure, the best-candidate structural break for a given station is the week with the highest *F*-statistic. Breaking in one of the four measures could occur for any number of reasons, but breaking in multiple markers in close proximity should provide a strong indication of adoption. Therefore, we classify a station as an AP adopter if it experiences a best-candidate break in at least two of four markers within a short time period, which we take to be 4 weeks, but is robust to alternative specifications. We find that approximately 20% of stations in our dataset experience best-candidate breaks in multiple markers within a 4-week window. The majority of these breaks occur in mid-2017, just as AP software supposedly became widely available in Germany. Adopting stations experience noticeably different trends in all four measures,

¹ Legal disclaimer: this paper analyses the impact of adoption of AP on competition strictly from an economic point of view. We base our understanding of the facts on publicly available data on prices from the German Market Transparency Unit for Fuels. To our knowledge, there is no direct evidence of anticompetitive behavior on the part of any algorithmic software firms or gasoline brands mentioned in this paper.

confirming that our data-driven approach for identifying adoption captures meaningful changes.

Having identified adopters, we next examine the impact of their adoption on retail prices and margins.² Although we control for time- and station-specific effects as well as time-varying market-level demographics, individual station adoption decisions may be correlated with station/time-specific unobservables (e.g., managerial skills, changing local market conditions). We provide evidence of selection bias and diverging outcomes between nonadopters and adopters before their adoption date that attenuate ordinary least squares (OLS) estimates to zero. We address this challenge by instrumenting for a station's adoption decision. Our main instrumental variable (IV) is the adoption decision by the station's *brand* (i.e., by brand headquarters [HQ]). As demonstrated by previous technology adoption episodes in retail gasoline, brands can facilitate adoption by their stations. Adopting brands provide support/subsidies/training to individual stations, reducing adoption costs.³ Brand-level decisions should not be correlated with individual station-specific unobservables, making this instrument valid. Since brand adoption decisions are also unobserved, we use a proxy as our instrument: the fraction of a brand's stations that adopt AP. If a large fraction of a brand's stations adopts, it is likely that the brand itself adopted and facilitated adoption by the stations.⁴

We find that following adoption, mean station-level prices and margins increase by approximately 1.3 cents per liter (cpl), or roughly 15% for margins.⁵ These estimates are similar in magnitude to claimed increases in gross profits achieved by stations employing AP software in Brazil and Denmark. Our findings provide evidence of the causal impact of adoption of AP software; however, it is not clear whether these higher margins can be attributed to changes in the degree of competition intensity rather than to factors such as an improvement in the ability to identify fluctuations in wholesale prices or to better forecast demand.

To isolate the effects of adoption on competition, we focus on the role of market structure. We begin by comparing adoption effects in monopoly and nonmonopoly markets. If adoption influences competition, its effects may be stronger for nonmonopolists than for monopolists. However,

² Previous studies on coordination in the retail gasoline market use margins (retail prices over wholesale prices) to evaluate competition (Clark and Houde 2013, 2014; Byrne and De Roos 2019), and theory papers on algorithmic competition also make clear predictions related to margins (Calvano et al. 2020; Brown and MacKay 2023).

³ Below we provide examples of other episodes of technology adoption in retail gasoline markets.

⁴ As an alternative instrument, we use an annual measure of broadband internet availability in the area around each station. See app. sec. G.4 (apps. B–G are available online) for additional discussion.

⁵ Estimates using alternative broadband availability IVs are qualitatively similar to the main estimates, although larger. See app. sec. G.4 for additional discussion of these results.

there is a lack of clear theoretical predictions on how AP should affect average prices in different market structures if its only function is to improve a seller's ability to tailor prices to time-varying demand or cost conditions. Therefore, we restrict attention to small oligopoly markets (with two or three stations) to hold market structure roughly fixed and perform a more direct test of the predictions of the literature studying the impact of AP. We compare market-level margins in markets where no stations adopted, where a subset of stations adopted, and where all stations adopted. In the first type, competition is between rule-based algorithms. In the second, it is between rule-based and AI-powered algorithms, while in the last, it is between only AI-powered algorithms. By comparing all three market types, we can learn about the effect of AP on competition.

We observe heterogeneity in outcomes based on market structure suggesting that AI-powered AP software may affect margins through competition. Adopting stations with no competitors in their local markets (i.e., monopolists) see no statistically significant change in mean margins or prices. In contrast, adopting stations with local competitors experience a statistically significant margin increase of 1.3 cpl.⁶ Our oligopoly market-level results indicate that, relative to markets where no station adopts, markets where all do see a margin increase of 3.2 cpl, or roughly 38%. Mean prices increase by 6 cpl. Markets where only a subset of stations adopts see no change. These results show that market-wide AP adoption raises margins, suggesting that algorithms soften competition. The magnitudes of margin increases are consistent with previous estimates of the effects of coordination in retail gasoline markets (Clark and Houde 2013, 2014; Byrne and De Roos 2019).

To provide further evidence of the impact on competition and to better understand the mechanism, we examine whether algorithms actively learn how not to compete (i.e., to tacitly collude) by testing the timing of price and margin changes. Updating algorithms operating in fluctuating markets should adjust slowly, as they learn and explore the state space and set of possible outcomes. As a result, convergence to stable strategies can take as long as several years. Asker, Fershtman, and Pakes (2021) show that less sophisticated asynchronous algorithms converge toward the monopoly price but take many periods to do so. Similarly, Calvano et al. (2020) suggest that it can take time for algorithms to train and converge to stable strategies, involving, for instance, punishment for rival price reductions. We find evidence consistent with these results. Margins start to increase only about a year after market-wide adoption, suggesting that algorithms in this market learn tacitly collusive strategies. We also examine behavior that emerges in markets where all stations adopt. In these

⁶ We find that the pricing behavior of adopting monopolists changes in ways that do not increase average daily prices and is consistent with improved ability to price discriminate.

markets, a station is more likely to respond to a rival's price decrease with an immediate decrease of its own. There is no comparable change in the propensity to respond to rival price increases. The timing of these effects is consistent with that of the price and margin increases. Altogether, these findings provide further evidence that adoption affects competition and suggest that algorithms learn that undercutting will not be profitable, since lower prices will be followed.

Our results have important policy implications. Globally, antitrust authorities are considering adjustments to their toolkits to address the challenges of the digital economy (Autorité de la Concurrence and Bundeskartellamt 2019; UK Digital Competition Expert Panel 2019). Currently, authorities expend substantial resources pursuing hard-core cartels on individual bases, possibly overlooking a broader set of collusion-facilitating devices that do not even require a conspiracy. AP may be one such mechanism. Communication via earnings calls is another (see Aryal, Ciliberto, and Leyden 2022). We provide further policy discussion along with some recommendations in section VIII.

The remainder of this paper is laid out as follows. Section II discusses relevant literature. Section III provides a background discussion and an overview of the German market. Section IV describes the data and methodology used to identify AP adopters. Section V displays results on the impacts of AP adoption on outcomes. In section VI, we provide evidence that results are driven by algorithms learning to tacitly collude. Section VII presents a series of robustness results. Finally, in section VIII, we present a brief policy discussion and some conclusions.

II. Related Literature

This paper is most closely related to the recent literature concerning the potential link between AP and collusion. Theoretical and experimental results remain ambiguous. Several papers have shown that when AP competition is modeled in a repeated game framework, collusive outcomes are possible under certain conditions (Salcedo 2015; Calvano et al. 2020; Klein 2021); however, others argue that improved price response to demand fluctuations may provide increased incentives for firm deviation from a collusive price (Miklós-Thal and Tucker 2019; O'Connor and Wilson 2021). Klein (2021) and Calvano et al. (2020) use computational experiments to study the effect of Q-learning algorithms on strategic behavior of competing firms. Both find that these repeated games will converge to collusive outcomes, including supracompetitive pricing and profits, as well as punishment of competitor deviation.⁷ Asker, Fershtman, and Pakes

⁷ Johnson, Rhodes, and Wildenbeest (2023) propose market design policies to disrupt algorithmic collusive strategies in platform settings.

(2021) find that the sophistication of an algorithm's design affects the extent to which prices increase above the competitive benchmark. While Miklós-Thal and Tucker (2019) find that improved demand prediction may lead to the possibility of collusion in markets where it is previously unsustainable, in other markets it may create incentives for deviation that were absent with less prediction capabilities. O'Connor and Wilson (2021) come to similar conclusions. Brown and MacKay (2023) develop a model where firms compete in pricing algorithms (rather than prices) and show that prices may increase even without collusion.⁸ Overall, there is little certainty as to whether algorithmic competition will lead to collusive outcomes in reality. There is, as far as we are aware, no empirical research regarding this question in the economics literature.⁹

The question as to whether the use of algorithms may result in coordinated behavior has been studied in fields outside economics, such as law and computer science. In computer science, Kaymak and Waltman (2006, 2008) and Moriyama (2007, 2008) indicate that reinforcement learning algorithms can result in cooperative outcomes; however, these outcomes are not always the most likely and are dependent on various specifications of the algorithm. Legal scholars generally voice more certainty that AP can lead to collusion. Ezrachi and Stucke (2015, 2016, 2017) and Mehra (2015) have expressed concern over this issue and its implications for competition policy.

We also relate to an extensive literature on retail gasoline markets. A number of papers have examined collusion, including Borenstein and Shepard (1996) as well as Slade (1987, 1992). More recently, Wang (2009), Erutku and Hildebrand (2010), Clark and Houde (2013, 2014), and Byrne and de Roos (2019) have all studied anticompetitive behavior in retail gasoline. There is a small set of papers looking at the German retail gasoline market (Dewenter and Schwalbe 2016; Boehnke 2017; Montag and Winter 2019; Cabral et al. 2021).

An associated literature studies the impact of technological advancements on price discrimination. A consequence of the rapid expansion of Big data and AI-driven analysis is that personalized pricing strategies

⁸ See also Lamba and Zhuk (2022). Harrington (2022) shows that outsourcing the development of pricing algorithms to profit-maximizing third party developers can also increase prices and reduce consumer welfare by making algorithms more sensitive to changes in demand.

⁹ Decarolis and Rovigatti (2021) find that common bidding intermediaries in online advertising markets lead to anticompetitive effects, reducing prices for bidders at the expense of the platform. Bidding is done through algorithms, which leads to regulatory concerns about multiple competitors in a market adopting the same pricing algorithm. Their findings suggest that algorithms could serve as hubs in a hub-and-spoke cartel (Garrod, Harrington, and Olczak 2021; Clark, Horstmann, and Houde 2024). The primary focus is on increasing intermediary concentration rather than on AP software behavior. Two recent working papers study the rise of automatic pricing tools at e-commerce sites and investigate whether they can facilitate collusion (see Wieting and Sapi 2021; Musolff 2022).

may become increasingly sophisticated. It is possible that more accurate determination of optimal personalized pricing can increase firm revenues (Shiller and Waldfogel 2011; Shiller 2020). Kehoe, Larsen, and Pastorino (2020) find that firm profit—and consumer surplus—may increase or decrease under personalized pricing, depending on consumer certainty regarding product tastes, and that total welfare is higher under discriminatory pricing in comparison to uniform pricing. Dubé and Misra (2023) show through experiments that personalized pricing improves firm profits and that a majority of consumers benefit. MacKay, Svartback, and Ekholm (2022) also show efficiency gains under dynamic pricing.

III. Background

A. *The German Retail Gasoline Market*

As in other retail gasoline markets around the world, a large fraction of stations in Germany have a brand affiliation.¹⁰ Aral and Shell are the largest, combining to make up over 25% of stations in Germany.¹¹ There are a number of other large brands with over 350 stations each: Esso, Total, Avia, Jet, Star, BFT, Agip, Raiffeisen, and Hem. Aral, Shell, Jet, BFT, Total, and Esso together account for 84% of fuel sales in the German retail gas market.¹²

Two features of the German market are important for our analysis. First, price transparency was instituted in August 2013 in response to concerns about high prices and tacit collusion by regulatory authorities.¹³ As part of this initiative, stations adjusting their price must report new prices in real time to the German Market Transparency Unit for Fuels (www.bundeskartellamt.de), which are then shared with consumer-facing information service providers and integrated into websites and mobile applications as well as into car GPS systems.¹⁴

Second, Shell introduced a price-matching guarantee in 2015. Each Shell station was required to match the lowest price of the 10 nearest

¹⁰ Our dataset does not specify which stations are vertically integrated and directly owned by the brands and which are owned by independent franchisees who enter into a licensing agreement in exchange for the brand name and some technical support. Both are common in retail gasoline markets (convenience.org).

¹¹ Detailed summary statistics of station numbers at the brand level are in sec. IV.

¹² The 2019 shares of fuel sales are as follows: Aral, 21%; Shell, 20%; BFT, 16%; JET, 10.5%; Total, 9.5%; and Esso, 7% (bft.de).

¹³ The purpose is to allow “motorists . . . to gain information on the current fuel prices and find the cheapest petrol station in their vicinity or along a specific route” and to “increase competition” (www.bundeskartellamt.de). Evidence on the effect of this policy on prices and margins in Germany is conflicting (Dewenter, Heimeshoff, and Lüth 2017; Montag and Winter 2019). See Luco (2019) for analysis of a similar program in Chile.

¹⁴ A full list of consumer facing data providers is available at https://www.bundeskartellamt.de/EN/Economicsectors/MineralOil/MTU-Fuels/mtufuels_node.html. We obtained our data from Tanker-König, one such provider.

stations within a 30-minute period for consumers with Shell loyalty cards. Dewenter and Schwalbe (2016) and Cabral et al. (2021) find that the policy led to retail price increases, attributed to an additional price jump incorporated into the daily price cycles that characterize Germany's retail gasoline market. Before the policy, prices were elevated and then gradually decreased starting around 8 a.m. before rising sharply again in the evening. After the policy was implemented, a price jump at noon followed by reversion emerged. Overall, stations featured considerable price variability throughout the day.¹⁵ We perform several robustness checks to confirm that Shell stations (or their direct competitors) are not driving the main results (see app. G.1).

Our paper takes this environment as given, and so we study the additional effects of AP software in a market with price transparency, daily price variation, and price matching.

B. Use of Algorithmic Pricing Software in Retail Gasoline Markets

1. History of Algorithmic Pricing in Retail Gasoline

Fuel retailers are typically not forthcoming about the pricing technologies they employ. AI-driven AP software providers are also mostly secretive about their customer base, and little is known about the structure of the market or the market shares of particular software providers. A *Wall Street Journal* article on the subject mentions certain firms, including the Danish company A2i Systems and Belgian company Kantify, as notable providers (wsj.com). Other firms, not listed in the article but prominently featured on the internet as algorithmic software providers, include Kalibrate (kalibrate.com), Revionics (revionics.com), and PDI (pdisoftware.com).

The use of AP software in European fuel retail markets began in the early 2010s. A2i sold their software to Danish fuel retail company OK Benzin in 2011 (a2isystems 2016a). However, the main penetration of machine learning and AI-based software appears to have happened in the mid-2010s, roughly coinciding with the publication of several newspaper articles about the subject in 2017 (wsj.com, cspdailynews.com).¹⁶ Kalibrate

¹⁵ Similar patterns have been documented in other markets. For instance, Wang (2009) provides evidence that in Australia prices fluctuated multiple times throughout the day prior to the implementation of a pricing reform in 2001. The same is true of some markets in the United States and Canada (see ctvnews.ca).

¹⁶ It is possible that providers sold AP software in Germany before 2016 (the start of our sample). We should not observe any structural breaks for stations that adopted before the start of our sample. This means that we would be labeling some adopters as nonadopters. If adopters have higher average margins than nonadopters, this would bias our station-level estimates toward zero.

began explicitly distinguishing between rule-based pricing and AP on its website in mid-2017 (kalibrate.com 2016, kalibrate.com 2017). A2i's software was tested in workshops with stations in the Netherlands and Belgium in 2015 (servicestationmagazine.be) and adopted by some Shell stations in the Netherlands by 2017 (wsj.com).

In Germany, a number of trade publications and news articles mention that AP software became available in 2017, noting in particular the introduction of A2i's software.¹⁷ The websites of some software providers also suggest that they became active around this time. Evidence of this introduction and of adoption activity in Germany is presented in appendix section B.1.

Promotional materials by retail gasoline AP software providers around the world make claims that stations using their pricing software outperform stations with human pricing agents. The Brazilian pricing start-up Aprix estimates that gas stations using its AI-based pricing software increased station gross profits by approximately 10% (towardsdatascience.com). A2i similarly estimated that its software could increase station profits by at least 5% (a2isystems 2016a).

2. How Does Algorithmic Pricing Software Work?

Software providers reveal little about their algorithms, but promotional materials describe the software as based on "machine learning" or "artificial intelligence," with references to "neural networks" and "deep learning" (kalibrate.in, insidebigdata.com; a2isystems 2016a). They are characterized as able to help station owners "master market volatility with AI-powered precision pricing and respond rapidly to market events and competitor changes" (kalibrate.com) and take advantage of "super-human expertise" (a2isystems 2016b). Additional promoted benefits include optimizing for long-term revenues and avoiding price wars (Subramanian 2016).

Providers stress the ability of algorithms to incorporate market conditions and variables such as own and competitor prices, sales volumes, costs, and weather and traffic events into their decision-making. A2i Systems provides more detail, outlining its algorithm in Derakhshan, Hammer, and Demazeau (2016).¹⁸ It is described as a multiagent system based on the interaction of two agents: a consumer and a gas station. Agent behavior is described by a belief-desire-intentions (BDI) model, a popular approach in computer science and information systems research. An agent's

¹⁷ In conversations with us, A2i claims that, contrary to statements in these advertising materials, they were never active in the German market.

¹⁸ This algorithm is based on earlier papers (Derakhshan, Hammer, and Lund 2006; Hammer et al. 2006). These papers look at interactions of children at a playground, with the goal of encouraging more physical activity.

beliefs, desires, and intentions roughly correspond to information, pay-offs, and actions/strategy in decision theory.¹⁹

A2i's algorithm works in three repeating steps. The first is observation, where the agent collects data from the environment and forms beliefs. As mentioned previously, these data include own prices, sales, traffic, and environmental factors. Competitor behavior is not explicitly modeled, but rival prices are included as inputs at this step. In the second step—learning—the gas station agent uses an artificial neural network to map inputs into outcomes.²⁰ The outcomes are not explicitly outlined in Derakhshan, Hammer, and Demazeau (2016) but likely correspond to sales, revenues, and/or profits.²¹ These are the desires/payoffs in the BDI model. The last step is adaptation, where the agent sets prices to achieve their desires/maximize the objective function.²²

An interpretation of AI-driven AP software is that it makes stations more sensitive to the state of the market. The *Wall Street Journal* presents a summary of its functioning, describing constant learning about the state of the market (wsj.com). For retail gas, this means collecting demand-related information (such as weather and traffic) that can change driving behavior and the probability of stopping for gas, cost-related information (such as crude oil price fluctuations), or other relevant information (such as competitor prices). This can now be done at high frequency by scraping websites (e.g., weather websites or Google maps). The algorithm then relates this information to outcomes and decides on the best price to set conditional on the state. Human or rule-based pricing agent operators can also collect information about what competitors and consumers do and set prices in response (time.com), but algorithms collect and process more information than any human could. Algorithms can also respond faster to changes in the state and to very subtle changes that humans might miss, consistent with evidence from hotels showing that human pricing agents exhibit more inertia and higher price adjustment costs than algorithms (Garcia, Tolvanen, and Wagner 2021).

¹⁹ Individual station owners can set different goals (such as market share maintenance) or constraints (such as minimum price). They can also change the goals over time or adjust them. However, substantial changes by station owners does not happen much in practice. One algorithmic software provider states that approximately 80%–90% of station owners do not customize or interfere with the default operations of the algorithm (kalibrate.com).

²⁰ This step also implicitly models consumer behavior, but this is not described.

²¹ In the earlier papers on children's playgrounds that form the basis of this algorithm, outcomes are categories that capture whether children are playing fast or slow, continuously or discontinuously, and so on (Derakhshan, Hammer, and Lund 2006).

²² Similarly, the Brazilian provider Aprix claims to “simulate the demand reaction for different price and market scenarios” by cycling through three stages: modeling station and consumer behavior, simulating (or mapping) the relationship between inputs (the state) and desired outputs (margins, profits, market shares), and optimization by setting station prices to reach maximum outputs conditional on the state. As with A2i, the algorithm continuously performs these stages and reoptimizes (towardsdatascience.com).

For further discussion about how the algorithms operate, see appendix section B.2.

C. Algorithmic Pricing Software Adoption

As in other cases of corporate technology adoption (e.g., Tucker 2008; Ryan and Tucker 2012), technology adoption in gasoline retail happens at two levels: at the brand HQ level and at the individual station level. Brands make big-picture decisions about technologies they would like their stations to use and provide stations with employee training, technical support, maintenance, and subsidies. Individual station owners make adoption decisions specific to their stations, potentially incurring substantial investment costs that are not necessarily fully subsidized by the brand.

An example is electronic payment system adoption in the 1990s. As with AP software, brands wanted their stations to adopt this technology, but some stations may have been reluctant because of the costs involved. As part of a brand-wide rollout of a contactless electronic payment system, in 1997 Exxon Mobil offered a \$1,000 rebate toward the \$17,000 installation fee (per pump).²³ Partial subsidies help explain staggered/delayed technology adoption in this market.²⁴ We look at the adoption of electronic payments from 1991 to 2001 by Canadian gasoline retail stations and document that it takes years after the first appearance of this technology for a substantial fraction of stations belonging to the five biggest brands in the market to adopt. Even after 10 years of availability, fewer than 50% of stations owned by leading brands adopted the technology (fig. F1; figs. B1–F1 are available online).

There is no reason to suspect that AP software adoption is different. Anecdotal evidence suggests that gasoline brands have entered into long-term strategic partnerships with AI pricing and analytics providers either directly or indirectly.²⁵ However, should a brand decide to adopt or enter into a partnership with an AI pricing software provider, its stations do not necessarily automatically and instantaneously adopt for a variety of reasons. Cloud-based AI pricing software may require substantial infrastructure

²³ See businessweek.com.

²⁴ We provide additional evidence for staggered technology adoption in the gasoline retail market in app. F.

²⁵ For example, in Denmark, A2i directly entered into a partnership with Danish retail fuel company OK Benzin (a2isystems.com). More indirectly, AI pricing software providers enter into partnerships with information technology companies that provide integrated services to brands. Tankstop's June 2017 issue mentions that A2i's services are supported by WEAT Electronic Data Service, a provider of cash-free payment systems and technical and logistical support for a number of petrol brands within Germany (weat.de). A2i also has a strategic partnership with Wincor Nixdorf, a retail technology company providing services such as point-of-sale terminals and self-checkout solutions (dieboldnixdorf.com).

investments, and not all station owners are in a position to incur these costs when the technology becomes available.²⁶ In Germany, many areas do not have access to stable high-speed internet connections.²⁷ Station operators also require training with the software to set its parameters and deal with potential errors.

IV. Data

This section provides a general description of the datasets used in our analysis. The Replication Package (Ershov et al. 2024) contains more details about data construction. The main dataset comes from the German Market Transparency Unit for Fuels and includes all price changes for the most commonly used fuel types (E5, E10, diesel) for approximately 14,500 German gas stations. For each station, the raw data also include location information (five-digit zip code, latitude and longitude coordinates) as well as an associated brand.²⁸ Our sample covers January 2016 to December 2018.²⁹ We focus on E5 fuel, which has over 80% market share in Germany (bdbe.de).³⁰

We also make use of regional wholesale fuel prices from Oil Market Report, a private independent German gasoline information provider, and we merge in annual regional demographics from Eurostat. We include data on total population, population density, median age, employment (as a share of total population), and regional gross domestic product (GDP). These data are at the Nomenclature of Territorial Units for Statistics 3 (NUTS3) level, which is frequently used by European Union surveys. A NUTS3 region is roughly equivalent to a US county and larger than a five-digit zip code. We also incorporate weather information from the German Meteorological Service (DWD) and oil price data from FirstRate Data. Finally, we collect data on local fixed-line broadband internet from the EU Commission's netBravo initiative (netBravo): whether the local

²⁶ For example, high-speed internet as well as high-speed internet-enabled point-of-sale terminals and pumps are likely required for the software to work. Equipment upgrades of this sort are expensive, costing thousands to tens of thousands of euros (mobiletransaction.org). Again, this is analogous to previous cases of technology adoption and upgrading decisions by gas station owners, including allowing for chip cards or automated payment at the pump (chicagotribune.com).

²⁷ Reports suggest that many German regions receive subpar services and speeds comparable to the old dial-up days (npr.org). We use broadband internet availability as an alternative instrument. See discussion in app. sec. G.4.

²⁸ We do not observe the ownership structure of the stations.

²⁹ Additional data exist for 2014–15, but we choose to start our sample 2 years after the beginning of the transparency initiative and 1 year after Shell's price matching policy described in sec. III.A. Results are robust to alternative samples (app. sec. G.1).

³⁰ Super E5 is an ethanol-based fuel with 5% ethanol and 95% unleaded petrol. We find similar results using E10 fuel and diesel, as reported in sec. VII.

TABLE 1
BRAND AND MARKET SUMMARY STATISTICS

Variable	Observations	Mean	SD	Minimum	Maximum	P25	P75
Stations per brand	232	59	224	2	2,256	3	18
Stations per market	3,957	4	2	1	19	2	5
Stations per five-digit zip code	5,488	3	2	1	16	1	3
Distance to nearest station (km)	14,565	1.42	1.77	0	17.19	.32	1.69

area around the gas station has widespread availability of 10 Mb/s internet in a given year.³¹ Unfortunately, we do not have access to sales data, so to confirm that there were no significant changes during our sample period, we study aggregate volumes obtained using snapshots from the Wayback Machine of the website <https://www.bdbe.de/daten/marktdaten-deutschland>.³² Throughout the time period, sales of E5 hover around 15 million tons (14.6 million in 2014, 15.0 million in 2015, 15.1 million in 2016, 15.0 million in 2017, 14.7 million in 2018).

A. *Station-Level Descriptive Statistics*

Table 1 presents summary statistics, including the number of stations per brand, the number of stations per market, and the distance between stations. Out of the 14,565 stations in our dataset, single operating stations account for approximately 7%. With our IV strategy, these stations are not part of our final estimating sample. The remaining stations are affiliated with brands.³³ There are 232 distinct brands in the data, of which 215 have between two and 100 stations and 17 have more than 100. The top five brands account for 46% of stations, and the 19 largest brands (those with more than 100 stations) account for 74% of total stations (10,720 stations total).

The market definition we use is based on the clustering algorithms developed in Carranza, Clark, and Houde (2015) and Lemus and Luco (2019). The algorithm is implemented using distances between station

³¹ We define 10 Mb/s to be widely available in an area if average speed tests in that area in that year exceed that speed. More details on the construction of this variable are in the Replication Package (Ershov et al. 2024).

³² See <https://web.archive.org/web/20181116121312/https://www.bdbe.de/daten/marktdaten-deutschland> and <https://web.archive.org/web/20190929194231/https://www.bdbe.de/daten/marktdaten-deutschland>.

³³ The dataset does not specify whether the stations are vertically integrated and directly owned by the brands or whether they are owned by independent franchisees who have entered into a licensing agreement in exchange for the brand name and some technical support. Both are common in retail gasoline markets (convenience.org).

pairs. Details are provided in appendix C. Using this approach, we find that there are 3,957 markets, of which 526 have a single station (monopoly markets), 789 have two stations (duopolies), and 879 have three stations (triopolies). The full distribution is presented in the appendix. The mean number of stations per market is around four. Only 60 markets have more than 10 stations.³⁴

B. Station/Month-Level Descriptive Statistics

Using prices in the raw data, we calculate a mean weekday (nonweekend or holiday) price from 7 a.m. to 9 p.m. for each station. To construct margins, we merge these with regional wholesale prices (average daily external prices in eight major German refinery and storage areas). We calculate the distance between each station and all refinery and storage areas and use wholesale prices from the nearest refinery.³⁵ Prior to subtracting wholesale price, we also subtract German value-added tax (VAT; 19%) from retail prices. We compute station-level daily margins and take the monthly mean for our station/month-level analysis. In addition to daily average prices, we also calculate prices at different points during the day for each station. For each station and weekday, we calculate the price at 9 a.m., noon, 5 p.m., and 7 p.m. Once again, we take a monthly mean for our station/month-level analysis.

Table 2 displays summary statistics at the station/month level for prices, margins, and regional demographics and weather. The mean price charged is 1.36 euros per liter, and the monthly mean margin earned by the average station over wholesale regional prices (after subtracting VAT) is 8.3 cpl. The average station is located in a fairly dense NUTS3 region, with population density of 760 persons per square kilometer. The median age of the population around a station is 46 years old, and 53% of the population is employed. Over 83% of gas station/month observations are for areas with widely available 10 Mb/s internet access. The weather data are collected daily from thousands of weather stations. We compute the average distance between each station and all local weather stations and use data from the nearest weather station. We include monthly means and standard deviations of temperature (in degrees Celsius) and precipitation (in millimeters).

³⁴ In app. sec. G.2, we also consider a market definition based on five-digit zip codes. In Europe, this is the most detailed zip code available. There are 5,488 five-digit zip codes in our data, of which 2,039 have a single station (are monopoly markets) and 1,286 have two stations (are duopolies).

³⁵ This is a standard approach in the retail gasoline literature. We may be understating retail margins if stations belong to vertically integrated retailers.

TABLE 2
STATION/MONTH SUMMARY STATISTICS

Variable	Observations	Mean	SD
Prices and margins:			
Mean monthly E5 price (EUR/liter)	448,221	1.362	.083
Mean monthly E5 margin (EUR/liter)	448,221	.083	.032
Regional demographics and weather controls:			
ln(total regional population)	448,221	12.419	.816
Regional population density (population/km ²)	448,221	758.238	1,022.41
Regional median population age	448,221	46.018	3.125
ln(regional GDP)	448,221	9.083	.976
Regional employment share (employed/population)	448,221	.527	.134
Mean temperature (°C)	448,221	10.417	6.87
SD temperature (°C)	448,221	3.079	.806
Mean precipitation (mm)	448,221	1.94	1.399
SD precipitation (mm)	448,221	3.603	2.605
Broadband availability:			
At least 10 Mb/s internet available dummy	443,752	.834	.371

C. *Identifying Algorithmic Pricing Adoption*

1. Station-Level Adoption

We do not have information on the algorithmic pricing software adoption of individual stations or brands. Our approach is to take advantage of the detailed price data to identify changes in station pricing technology, since we see price changes at 1-minute intervals. As discussed in section III.B, algorithms use machine learning to optimize prices conditional on a state that includes variables such as competitors' prices, weather conditions, and traffic. Changes in a station's price responsiveness to the state should indicate the adoption of AP software. We consider the following four variables that capture a gas station's responsiveness to the state:

1. Number of price changes made in a day: we calculate the number of times each gas station changes their price in each nonholiday weekday. We then average across each week.
2. Average response time to a rival's price change: we define a rival to station i as the nearest station j that belongs to a different brand and is within 1 kilometer of station i . After each price change by station j in each nonholiday weekday, we calculate the average time in minutes it takes station i to respond. We then average across each week.
3. Responsiveness of a station's prices to crude oil price shocks: using data from FirstRate Data, we observe an intraday time series for crude oil prices. In each nonholiday weekday, we separate fluctuations in crude oil prices from the moving average. We define a

crude oil price shock as large deviations from the moving average.³⁶ We define a response to a crude oil price shock as a price change within 5 minutes of the shock. We calculate the number of shocks for each week and the number of responses and response probability for each station and week.

4. Responsiveness of a station's prices to local weather shocks: using data from the DWD, we observe a high-frequency time series of local air temperature around each station.³⁷ We separate fluctuations in temperature from the moving average. We define a local weather shock as a large deviation from the moving average³⁸ and a response to a local weather shock as a price change within 5 minutes of the shock. We calculate the number of local shocks, number of responses, and response probability for each station and week.

Similar measures have also been used previously in the literature to identify heterogeneity in pricing technology (Chen, Mislove, and Wilson 2016; Aparicio, Metzman, and Rigobon 2021; Brown and MacKay 2023).³⁹

Formally, we look for structural breaks in these measures using Quandt likelihood ratio (QLR) tests (Quandt 1960; Andrews 1993). This method tests for the best-candidate structural break in a time series measure for each period in some interval of time and takes the largest resulting test statistic. It is useful when an exact break date is unknown and has been used in previous work involving collusive behavior (Harrington 2008; Clark and Houde 2014; Boswijk, Bun, and Schinkel 2018; Byrne and de Roos 2019). We conduct a QLR test for each station and for each measure. Further details can be found in appendix D.

Appendix A shows the distribution of structural breaks for each measure.⁴⁰ We find a large number of statistically significant breaks in the

³⁶ Defined as deviations from the moving average above the 90th percentile of all deviations in a given year-month, helping to account for changing volatility of oil prices over time.

³⁷ We also observe local precipitation. Fluctuations and shocks in precipitation are generally highly correlated with shocks in temperature, but there are some areas and time periods that are drier and so have no precipitation and no fluctuations. Variation in air temperature exists for all stations throughout our sample period.

³⁸ Defined as deviations from the moving average above the 90th percentile of all deviations in a given year-month.

³⁹ Brown and MacKay (2023) use the number of price changes by retailers in a given period and the speed of reaction to identify new pricing technology. Similarly, Aparicio, Metzman, and Rigobon (2021) document a higher frequency of price changes by online retailers who use AP. Chen, Mislove, and Wilson (2016) also identify AP users in Amazon Marketplace by measuring the correlation of user pricing with certain target prices, such as the lowest price of that given product in the marketplace.

⁴⁰ A concern is that other breaks may occur at different dates if we considered F -statistics that are not the maximum but close to it. We find that F -statistic distributions are generally unimodal and that stations do not have significantly different dates that may be identified as breaks. Examples of F -statistics distributions are in fig. D3.

data. Nearly 50% of best-candidate breaks in the number of price changes are in the spring of 2017, when we believe AP technology became available. Similarly, 40% of best-candidate breaks in the responsiveness to local weather shocks, 20% in rival response time, and nearly 20% in responsiveness to oil price shocks happen around that time.⁴¹

We also find that the breaks capture quantitatively important changes in pricing behavior. On average, stations without structural breaks in the number of price changes adjust their prices approximately five times per day, or roughly once every 3 hours, assuming average opening hours from 7 a.m. to 9 p.m.⁴² Stations with structural breaks change their prices approximately eight times per day during the sample period, or approximately once every hour and a half (9.3 times per day in 2018).⁴³

For response time to rivals, the average for stations without breaks is over 84 minutes compared with 50 minutes for stations with a break.⁴⁴ This is at least as fast as the responsiveness identified in Aparicio, Metzman, and Rigobon (2021) and Brown and MacKay (2023), who find that even firms with the most sophisticated pricing technology (e.g., Amazon) do not respond to competitors' price changes for several hours (on average).⁴⁵ We identify approximately 3.7 weather shocks per week and 0.9 responses for stations with structural breaks after their break date compared with 0.5 for stations without breaks. Similarly, we identify an average of 12 oil price shocks per week and find that stations with breaks respond 1.2 times per week after their break date compared with once for a station without a structural break.⁴⁶

⁴¹ Since variation in responsiveness to oil price shocks appears to be less clear-cut in the data as compared with the other three measures, we also test a definition of adoption that excludes it. Our main results hold.

⁴² The number of price changes increases slightly throughout the sample period, from 4.8 to 5.3.

⁴³ This frequency of price changes is similar to the most rapid (hourly) average price change frequencies identified in online markets by Aparicio, Metzman, and Rigobon (2021) and Brown and MacKay (2023). See also Musolf (2022), who finds that sellers on Amazon using third-party repricers adjust prices every 0.29 hours.

⁴⁴ The difference in raw averages is larger than the difference in fig. D2 since fig. D2 accounts for station fixed effects that absorb some of the heterogeneity. We should also note that average response speed hides substantial heterogeneity. We show a substantial increase in very rapid (5 minute) responsiveness to competitors' price changes in duopoly and triopoly markets in sec. VI, consistent with the description of retail gasoline algorithms in sec. III.B.

⁴⁵ Online pricing in multiproduct markets may be more complex than offline pricing in the relatively homogenous retail gasoline market, but online retailers should have data that are at least as good as German gas station data. Online retailers can easily and continuously scrape competitor prices and demand proxies, such as sales ranks. Online technology should be at least as advanced as offline technology, so this appears to be the frontier of AP in retail markets in terms of the number of price changes and the average speed of competitive response.

⁴⁶ Since variation in responsiveness to oil price shocks appears to be less clear-cut in the data as compared with the other three measures, we also test a definition of adoption that excludes it. Our main results hold.

Figure D2 confirms these effects, presenting estimates from a station/month-level regression of $y_{it} = \Sigma_i \gamma_i^1 D_t + \Sigma_i \gamma_i^2 D_t \times \text{Break}_i + \delta_i + \epsilon_{it}$, where D_t is a dummy for month t , δ_i are a set of station fixed effects, and Break_i is a dummy equal to 1 for stations that experienced a structural break. γ_i^2 represents the average difference in outcomes between stations with and without best-candidate breaks in month t . The figure shows that for each measure, stations that break behave differently than those that do not: 20% more changes, 10% faster response to rivals' price changes, 20% more frequent response to weather shocks, and 5% more frequent response to oil shocks. Importantly, the changes between stations with and without breaks all appear very rapidly around the middle of 2017.

Classification.—Many factors may influence a single measure of pricing behavior on its own, but breaking in multiple markers in close proximity should provide a strong indication of an actual change in pricing technology, which in our case is the adoption of AP pricing. We label a station as an adopter of AP software if it experiences best-candidate structural breaks in at least two measures of pricing behavior within 4 weeks.⁴⁷ Our results are robust to stricter definitions of adoption.⁴⁸ We classify 2,728 stations as adopters. Figure 1 shows the distribution of the average break date for all adopters, defined as the average year-week between best-candidate break dates of the two or more measures in which a station experiences a significant break.⁴⁹ Over 50% of these average break dates occur in the middle of 2017, consistent with the supposed increased availability of AP software at this time in Germany (see sec. III.C).⁵⁰

The stations we classify as adopters show meaningful differences in their pricing behavior compared with stations without best-candidate structural breaks and stations with best-candidate structural breaks that are not

⁴⁷ Any combination of two measures will result in a station being classified as an adopter.

⁴⁸ In app. sec. G.3, we require stations to experience best-candidate breaks in at least two of the four measures within 2 weeks. We also include an additional definition that labels stations as adopters only if they experience multiple best-candidate breaks in two out of three measures (excluding rival response time and oil shock responsiveness), if they experience best-candidate breaks in Diesel, or if they experience best-candidate breaks in both E5 and Diesel within 4 weeks.

In app. sec. D.5, we present information on the break combinations across the four measures.

⁴⁹ This is a conservative approach. We may be missing some adopters because of either measurement errors in our markers or other signs of adoption that we did not consider. As a result, some adopters are classified as nonadopters, biasing our station-level estimates toward zero and understating the true effects of adoption.

⁵⁰ A possible concern with our classification is that nonadopting stations may be mistakenly labeled adopters if response to an adopting rival's pricing makes them behave as though they also adopted. This is not a regular occurrence. We observe a large number of duopoly markets where one station is classified as an adopter and not its competitor. Among 717 duopoly markets with full data in December 2018, 544 had no adopters, 142 had at most one, and 31 had two. More generally, fig. D5 shows the distribution of adoption shares in markets with more than one adopting station in December 2018 (the last month in our data). There are 1,700 such markets out of a total of more than 4,000 and relatively few where adoption shares are higher than 50%.

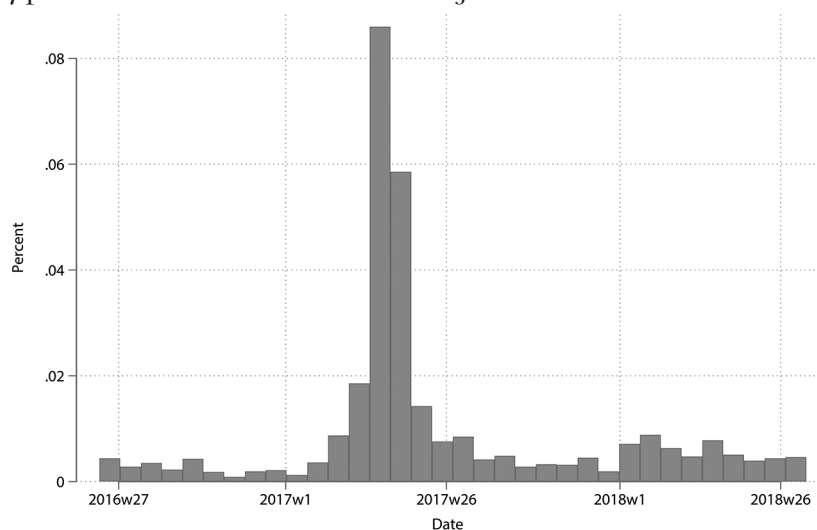


FIG. 1.—Frequency of average break date for measures breaking within 4 weeks (2,728 stations). This histogram shows the distribution of dates at which stations are labeled as adopters. We define an adoption date as the average year-week between best-candidate break dates of the two or more measures in which a station experiences a significant break.

classified as adopters. Figure 2 compares outcomes between adopter and nonadopter stations throughout our sample period.

We also find *ex ante* average differences in local demographic characteristics and local markets between stations labeled as adopters and those that are not.⁵¹ In table 3, we find statistically significant differences in market characteristics between adopter and nonadopter stations before any adoption takes place (in 2016). Adopter stations are located in denser areas with different demographic profiles. They also face more competition, suggesting that adoption decisions may be endogenous, with stations choosing to adopt in response to observable and unobservable market conditions.

2. Brand-Level Adoption

We do not observe whether a brand entered into a strategic partnership with an AP software provider. However, we can use findings from the station-level classification to infer brand-level adoption. We use a probabilistic definition, computing the probability that a brand adopted by time t as the percentage of a brand's stations classified as adopters by t . This captures underlying brand-level decisions. As described in section III.C, brand-level decisions should facilitate the adoption by individual stations. A

⁵¹ We do not observe individual station characteristics.

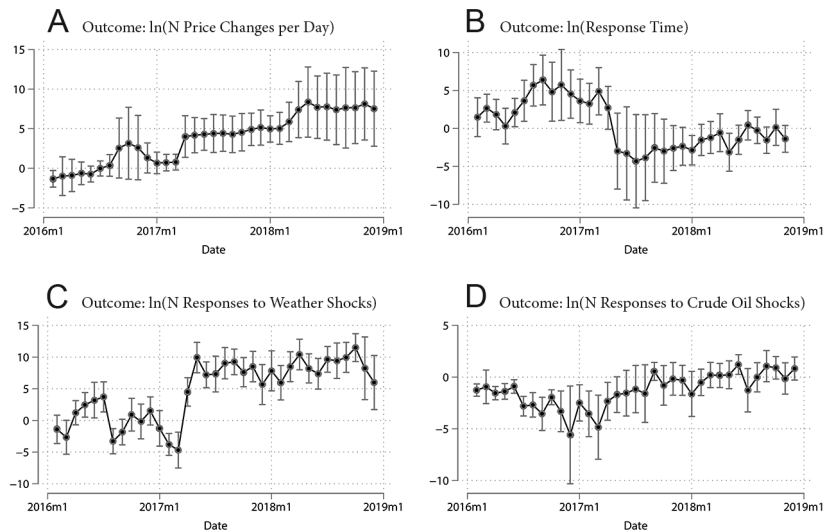


FIG. 2.—Percent difference between adopters and nonadopters. Each panel shows γ_t^2 estimates and their 95% confidence intervals from a station/month-level regression of $y_{it} = \sum_i \gamma_i^1 D_i + \sum_i \gamma_i^2 D_i \times \text{Adopter}_i + \delta_i + \epsilon_{it}$, where D_i is a dummy for month t , δ_i are a set of station fixed effects, and Adopter_i is a dummy equal to 1 for stations that are labeled as adopters. γ_t^2 represents the average difference in outcome y in month t between a station that is labeled as an adopter of AP and a station that is not labeled as an adopter. The outcome in A is the natural log of the average number of daily price changes a station has. The outcome in B is the natural log of the average response time to a rival's price change. The outcome in C is the natural log of the average number of station responses to a weather shock. The outcome in D is the natural log of the average number of station responses to a crude oil price shock. Additional controls in that regression include the number of competitors in the market and the average number of competitors' price changes.

brand for which a small percentage of stations adopted by t is unlikely to be an adopter at t , while a brand for which a large percentage of stations adopted is more likely to be an adopter.⁵²

Figure 3 shows the evolution of the share of a brand's adopting stations throughout our sample period. Figure 3A displays results for the top five

⁵² Alternative definitions could classify a brand as an adopter if any one of its stations is classified as adopting or only after all of its stations are. These alternatives do not reflect technology adoption in this market: brand adoption is not a necessary condition for station adoption. Many providers of AP software cater to small or medium enterprises (e.g., prisync.com or computera.net). A2i's 2017 advertisements target individual station owners and emphasize that all stations, regardless of brand, can adopt their technology. Defining a brand as an adopter if any one of its stations is classified as an adopter would be sensitive to outliers and amplify noise from our station-level adoption measure. Defining a brand as an adopter only if all of its stations adopt is inconsistent with the history of technology adoption in the market. As explained in sec. III.C, brand subsidies to stations for technology adoption are often incomplete, and technology adoption is highly staggered. In app. F, we show that it took years for a substantial share of gasoline stations belonging to top brands to adopt electronic payments in the 1990s.

TABLE 3
ADOPTER AND NONADOPTER STATION
CHARACTERISTICS IN 2016

Outcome	Will Station <i>i</i> Adopt AP?
Population density	.00003*** (.00001)
ln(population)	.00443 (.03513)
Median population age	.00707*** (.00211)
Employment share	.09257 (.07782)
ln(region GDP)	.00056 (.03241)
Competitors in market	.00297* (.00165)
Observations	165,810

NOTE.—The sample for this regression includes gas station/month observations from January 2016 to December 2016. The outcome is a dummy variable equal to 1 if the station will eventually be labeled as an adopter in 2017 or 2018 and zero otherwise. Population density, ln(population), median population age, employment share, and ln(regional GDP) are all computed at the NUTS3/year level. “Competitors in market” is equal to the number of other stations present in the market of station *i* in month *t*. We include month fixed effects. Standard errors (in parentheses) are clustered at the market level.

* $p < .10$.
*** $p < .01$.

brands (by station count). Adoption happens at a staggered rate that varies across brands. All brands experience an increase in adoption around early/mid-2017, likely reflecting the increased availability of the technology. Aral is an early adopter, with nearly 30% of its stations adopting by mid-2017. Total’s and Avia’s adoption rates increase at a steadier (albeit slower) pace compared with other brands. The heterogeneity in adoption rates across brands suggests that there is a brand-specific component to AP adoption, possibly reflecting that some brands were more likely to support the new technology. None of the top five have adoption rates over 40% by the end of the period. The share of adopters for smaller brands is typically lower and occurs later than for the large brands: the mean adopter share for top five brands is 21% by December 2018, while for brands outside the top five it is 12%.⁵³ This likely reflects the better support that larger brands can provide to their stations, lowering adoption costs. Figure 3*B* presents the mean for all brands (solid line) and

⁵³ Seven percent of single-station independents are labeled as adopters by the end of 2018.

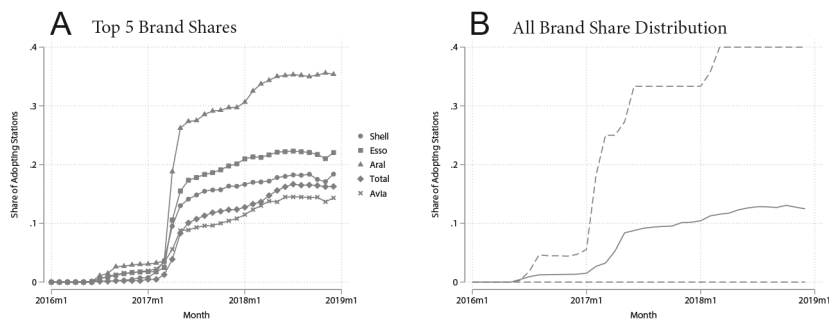


FIG. 3.—Share of brand stations that are AP adopters. *A* shows the share of brand stations that are AP adopters in each month for the top five brands in our data (by count of stations). *B* shows the distribution of brand adoption shares for all 258 brands. The solid line in *B* shows the mean brand adoption rate, while the area between the dashed lines shows the distribution of brand adoption shares between the 5th and 95th percentiles (for each month, we calculate the 5th and 95th percentile brand adoption share).

the area captured by the 5th and 95th percentiles. The difference between the 5th and 95th percentile grows over time, suggesting that heterogeneity in within-brand adoption rates is growing.

The pattern in figure 3 can be compared with the observed timing of electronic payment adoption at Canadian gasoline stations in the 1990s (see fig. F1). Despite differences in time, geography, and technology, we also see a staggered pattern of technology adoption that appears to be highly brand specific, suggesting that our AP classification captures technology adoption.⁵⁴

V. Results: Effects of AI Adoption

This section presents our estimates of the effects of AP adoption on prices and margins in the German retail gasoline market.

A. Impact of Adoption on Station Outcomes

1. OLS Estimation and Results

Our objective is to capture the effects of station i 's adoption of AP on average daily margins (above regional wholesale prices) and prices in period t . We use a station/month specification where we calculate average

⁵⁴ In app. sec. B.1, we also provide an example of staggered rollout of AP at a brand. According to news articles, Lekkerland—a company that operates gas station convenience stores in Germany and teams up with brands/stations—adopted dynamic pricing in its stores around 2017. The articles mention explicitly two stores with attached gas stations that were the test cases for the introduction of dynamic pricing technology in 2017.

monthly daily outcomes and characteristics for each station in month t ($t \in \{1, 2, \dots, T\}$). Our OLS specification is as follows:

$$y_{it} = \alpha_i + \alpha_t + \beta(\text{Adopter} \times \text{Postadoption})_{it} + \gamma X_{it} + \epsilon_{it}, \tag{1}$$

where y_{it} is the outcome variable for station i in time t ; α_i and α_t are station and time fixed effects, respectively; and $(\text{Adopter} \times \text{Postadoption})_{it}$ is a dummy variable equal to 1 if station i has adopted AP before period t and 0 otherwise. X_{it} are time-varying station-specific controls (local demographics and weather). X_{it} also includes the number of other gas stations that are in the same market as station i . The key coefficient in this regression is β , which captures the effect of AI adoption on y_{it} . Columns 1 and 2 in table 4 present the main average OLS station-level estimates. These show

TABLE 4
OLS STATION-LEVEL ESTIMATES

Outcome	Mean Margin (1)	Mean Price (2)	Mean Margin (3)	Mean Price (4)
Adopter $\times$ postadoption	.001** (.000)	.001** (.000)	.001*** (.000)	.001** (.000)
Adopter $\times$ 1–6 months preadoption			.000 (.000)	.000 (.000)
Adopter $\times$ 7–12 months preadoption			.002*** (.001)	.002*** (.001)
Adopter $\times$ 13+ months preadoption			.000 (.001)	.003*** (.001)
Nonadopter mean outcome	.0821	1.361	.0821	1.361
Station fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	478,172	478,172	309,280	309,280

NOTE.—In cols. 1 and 2, sample is gas station/month observations from January 2016 to December 2018. In cols. 3 and 4, sample includes only stations with a history of more than 12 months in the data. Margins are computed above wholesale gasoline prices at a regional terminal nearest to station i . Mean margin/price is the monthly average pump price for station i in month t . “Adopter $\times$ postadoption” is a dummy equal to 1 in month t if the gas station experienced a structural break in at least two of four relevant measures in any previous month $\{1, \dots, t - 1\}$. “Adopter $\times$ X months postadoption” is a dummy equal to 1 in month t if the gas station experienced a structural break in at least two of four relevant measures X months prior to month t . Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the number of stations belonging to station i ’s brand in month t for the number of competitors in the market and for the number of adopting competitors in the market. Weather controls include the mean and standard deviation of monthly temperature and precipitation near station i in month t . Standard errors (in parentheses) are clustered at the market level.

** $p < .05$.
*** $p < .01$.

that the adoption of AP increases average margins and prices by approximately 0.1 cpl.

We are concerned that OLS estimates are biased because of endogeneity. The OLS specification assumes that adoption is exogenous and as good as random (conditional on observables). Despite the inclusion of fixed effects and a rich set of station-level observables, this is likely not the case. AP adoption could be correlated with unobservable time-varying station characteristics (ϵ_{it}). The adoption of any new technology but especially of new pricing technology is an important and potentially costly decision with long-term consequences. Adopters are going to be stations that expect limited profits if they do not adopt the new software and that can afford to make the investment. This would mean that stations that have had better unobservable shocks in the past and that expect worse future unobservable shocks will be more likely to adopt: such patterns in the unobservables would generate negative correlation between the adoption decision and the ϵ_{it} shocks. Such stations would also have different market outcomes. This would invalidate a difference-in-differences (or event study) research design and attenuate estimated adoption effects toward zero.

Table 3 shows that adopter and nonadopter stations are very different in their local market demographics and in their competitive environment. They are also likely to be different in their unobservable characteristics. We provide further evidence of endogeneity in the OLS regressions using a formal test of parallel trends between adopters and nonadopters before adoption.⁵⁵ We estimate the following specification:

$$\begin{aligned} y_{it} = & \alpha_i + \alpha_t + \beta_1(\text{Adopter} \times \text{Postadoption})_{it} \\ & + \beta_2(\text{Adopter} \times 1\text{--}6 \text{ Months Pre})_{it} \\ & + \beta_3(\text{Adopter} \times 7\text{--}12 \text{ Months Pre})_{it} \\ & + \beta_4(\text{Adopter} \times 13 + \text{Months Pre})_{it} + \gamma X_{it} + \epsilon_{it}, \end{aligned} \quad (2)$$

where the key coefficients to estimate are the time varying β s, which represent the differences between adopter and nonadopter stations at various times. For example, $(\text{Adopter} \times 7\text{--}12 \text{ Months Pre})_{it}$ is a dummy equal to 1 for adopter stations 7–12 months before their actual adoption. The baseline period for each adopter station i is the month immediately before adoption.

Columns 3 and 4 in table 4 present the time-varying β coefficients and reveal statistically significant differences in mean prices and margins

⁵⁵ Such endogeneity would be mitigated by a flat specification where we do not consider time-varying adoption but simply calculate average outcomes and characteristics for adopters and nonadopters before and after the middle of 2017 ($t \in \{\text{before mid-2017, after mid-2017}\}$). However, even this specification would be subject to a downward bias if time-varying outcomes are correlated with time-varying shocks. See additional discussion in app. sec. E.1.

between adopter and nonadopter stations prior to adoption. Adopters had margins that were 0.2 cpl higher than for comparable nonadopters 7–12 months before they adopted. Similarly, adopters had prices that were 0.2 cpl higher than for nonadopters 7–12 months before adoption and 0.3 cpl a year before adoption. This suggests that the parallel trends assumption does not hold in our setting, invalidating a difference-in-differences/event study–based research design. These results also confirm the intuition described above: adoption of AP technology is a strategic decision made by stations that can afford to adopt and that expect limited profits if they do not. These stations likely had better unobservable shocks (and higher margins/prices) some time before adoption and may expect worse shocks in the future.

2. IV Estimation and Results

Since we are not able to use an event study research design, we turn to an IV approach to identify the causal effect of AP adoption on mean margins and prices. We need to instrument for $(\text{Adopter} \times \text{Postadoption})_{it}$. Our instrument should be correlated with an individual station's adoption decision but should not be affected by station-specific unobservable shocks. We propose brand HQ-level adoption as an instrument.⁵⁶ As explained in section V.A.1, we measure brand-level adoption by computing the share of stations belonging to each brand that have been identified as AP adopters by month t . For station i at time t , our IV is the share of stations in station i 's brand that adopted AP by time t . We exclude station i from this share.

The intuition behind this instrument is similar to the commonly used Hausman-Nevo instruments (e.g., Dubois and Lasio 2018). These instruments are valid if they appropriately recover common cost shocks across groups of observations, for example, by using prices from nearby observations as an instrument for own prices.⁵⁷ In our case, adoption costs should be correlated for stations within a brand because of the aforementioned

⁵⁶ As a robustness check, we propose an alternative set of instruments: the availability of broadband internet in the local area around a gas station. As with brand HQ-level adoption, the availability of broadband internet should have an effect on a station's decision to adopt AP software. Most AP software are cloud based and require constant downloading and uploading of information. Without high-speed internet, adoption of such software is not particularly useful for a station. However, the availability of broadband internet in the region should be uncorrelated with station unobservables after conditioning on observable local characteristics. Our estimates with these IVs are qualitatively similar to our main estimates. See table G6 (tables D1–G11 are available online) for results and app. sec. G.4 for additional discussion. We also test a placebo IV that uses the brand HQ-level adoption decision by a random brand (not the brand of station i) as an instrument and find null effects. Additional discussion is also in app. sec. G.4.

⁵⁷ Dubois and Lasio (2018) effectively use the prices of pharmaceutical molecule combinations in Germany, Italy, Spain, and the United Kingdom as an instrument for the prices of the same molecule combination in France.

brand subsidies for technology adoption. Brand-level decisions likely influence the adoption decisions of individual stations (see sec. III.C for additional discussion). Brands provide individual stations with employee training, technical support, and maintenance (convenience.org). This happens for both chain-operated stations as well as for more independent lessees. For previous waves of technology adoption (such as electronic payments), brands also directly subsidized some costs associated with required station upgrades. This support is important for drastic technical changes, such as AP adoption. At the same time, brand-level decisions should not be influenced by station level-specific demand or supply conditions.⁵⁸

Station-level IV estimates are presented in table 5. Column 1 shows the first stage of the IV regression. The first stage is strong, with an *F*-statistic of 35. A 10% increase in the number of other stations affiliated with station *i*'s brand (excluding *i*) that adopt by period *t* increases the probability that *i* adopts by *t* by 66%, consistent with intuition that adoption of AP is at least in part a brand-level decision. Columns 2 and 3 of table 5 show two-stage least squares (2SLS) estimates with margin and price outcomes, and columns 4 and 5 show the reduced-form estimates. Column 2 shows that mean margins increase by 1.2 cpl on average after AP adoption, or about 15% relative to the average nonadopter margin of 8.2 cpl.⁵⁹ Prices also increase by 1.2 cpl after adoption. Reduced-form estimates confirm that there is a direct positive correlation between the instrument and the main outcomes.

The 2SLS estimates are approximately 10 times larger than OLS estimates. One potential reason for this difference is measurement error in our adoption classification. Indeed, if we weight observations by the inverse of the noise (uncertainty) created by our structural break tests, we obtain estimates that are approximately three times our baseline unweighted OLS estimates, bridging part—but not all—of the gap between the OLS and IV results and indicating that measurement error is responsible for some of the downward attenuation of the OLS results.⁶⁰ The bulk of the difference can be attributed to the fact that adopters will be stations that expect limited profits if they do not adopt the new software and that can afford to make the investment (stations that have had better unobservable shocks in the past and expect worse future unobservable

⁵⁸ Table D1 shows that, conditional on brand size, brand adoption shares are uncorrelated with market characteristics. We also test a placebo IV that uses the brand HQ-level adoption decision by a random brand (not the brand of station *i*) as an instrument and find null effects. These findings make sense if the brand-level IV actually recovers each brand's costs rather than some other time-varying common cost shocks. Additional discussion is in app. sec. G.4.

⁵⁹ The 2SLS regressions using alternative instruments based on broadband availability and quality also show that mean margins and mean prices increase after adoption (see table G6). See app. sec. G.4 for additional discussion.

⁶⁰ See table E2.

TABLE 5
IV STATION-LEVEL ESTIMATES

OUTCOME	FIRST-STAGE ADOPTER (1)	2SLS		REDUCED FORM	
		Mean Margin (2)	Mean Price (3)	Mean Margin (4)	Mean Price (5)
Adopter × postadoption		.012*** (.002)	.012*** (.002)		
Share brand adopters	.660*** (.041)			.008*** (.001)	.008*** (.001)
Nonadopter mean outcome		.0821	1.361	.0821	1.361
Station fixed effects	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes	Yes
Observations	448,221	448,221	448,221	448,221	448,221

NOTE.—Sample is gas station/month observations from January 2016 to December 2018. Margins are computed above wholesale gasoline prices at a regional terminal nearest to station i . Mean margin/price is the monthly average pump price for station i in month t . Margins are computed above wholesale gasoline prices at a regional terminal nearest to station i . Mean margin/price is the monthly average pump price for station i in month t . “Adopter × postadoption” is a dummy equal to 1 in month t if the gas station experienced a structural break in at least two of four relevant measures in any previous month $\{1, \dots, t - 1\}$. “Share brand adopters” is the excluded instrument used in the 2SLS regression. It is equal to the share of stations that belong to the brand of station i that adopted by period t (excluding i). Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the number of stations belonging to station i ’s brand in month t for the number of competitors in the market and for the number of adopting competitors in the market. Weather controls include the mean and standard deviation of monthly temperature and precipitation near station i in month t . Standard errors (in parentheses) are clustered at the market level.

*** $p < .01$.

shocks) since such patterns in the unobservables would generate negative correlation between the adoption decision and the ϵ_{it} shocks. Anecdotal evidence supports this: stations struggling to maintain high margins benefit most from adoption.⁶¹ It is also the case that OLS is estimating the average treatment effect over the entire population, while the IV is estimating

⁶¹ See towardsdatascience.com, which describes adoption motives in Brazil, pointing to retail margins restrained by both sides of the supply chain. The upstream segment reduces station margins through increased fuel purchase costs, while at the same time consumers in Brazil were facing tighter budget and becoming more price sensitive, making it difficult for stations to pass on cost increases. AI-driven AP can help stations “survive in the new highly competitive environment, turning the threat into an opportunity.” Another article quotes the president of retailer operating in California: “Maybe the manager hasn’t done a price survey yet, or we need them to update it because the markets are moving intraday rapidly during the hurricanes, and we need to do price surveys more than once a day because we are struggling to keep up with what is happening. . . . That’s the reason why AI is interesting: Is there a way for us to get information faster so we can react faster to changes in the market?” See csdailynews.com.

the local average treatment effect. That is, the instrument affects the behavior of a subgroup of stations for whom the returns to adoption are greater than average or, put differently, for the stations whose choice of treatment was affected by the instrument. The full population used in the OLS includes (1) stations that could never adopt because the fixed costs of doing so are prohibitive, (2) stations for which adoption was almost automatic because they were heavily subsidized by the brand, and (3) stations in between. The IV zooms in on the stations in between, estimating a local treatment effect that is naturally much larger than the OLS effect.

Reassuringly, the magnitude of the 2SLS effects is in line with estimates of the effects of AP software on gas station profitability released by software providers. The Brazilian pricing start-up Aprix estimates that gas stations using its AI-based pricing software increased their gross profits by approximately 10% (towardsdatascience.com). A2i similarly estimated that its software could increase station profits by at least 5% (a2isystems.com).

Moreover, we confirm that our IV approach resolves concerns about diverging parallel pretrends between adopters and nonadopters. First, we verify parallel trends in the outcome for the instrument, testing for parallel trends in the reduced form. We do this by estimating the same regression as equation (2) but with the instrument in place of the endogenous treatment variable:

$$\begin{aligned}
 y_{it} = & \alpha_i + \alpha_t + \beta_1(\text{Share Brand Adopters} \times \text{Postadoption})_{it} \\
 & + \beta_2(\text{Share Brand Adopters} \times \text{1–6 Months Pre})_{it} \\
 & + \beta_3(\text{Share Brand Adopters} \times \text{7–12 Months Pre})_{it} \\
 & + \beta_4(\text{Share Brand Adopters} \times \text{13+ Months Pre})_{it} + \gamma X_{it} + \epsilon_{it}.
 \end{aligned} \tag{3}$$

As above, the baseline period is a month before adoption for adopting stations.

Estimates from this regression are presented in table 6, and they show that there is no correlation between the outcome variables and the instrument interacted with preadoption time dummies, suggesting that parallel trends in the instrument hold. We also estimate an IV version of equation (2), where we instrument for each lead variable with an appropriately constructed IV. For example, the variable $(\text{Adopter} \times \text{13+ Months Pre})_{it}$ is instrumented with $(\text{Share Brand Adopters} \times \text{13+ Months Pre})_{it}$. The 2SLS estimates from this regression are also in table 6, and they similarly show that there is no correlation between instrumented leads of the treatment variable and the outcomes. These results suggest that our instrument helps to effectively correct for the endogeneity between station-level adoption and other station-specific unobservable factors that can affect station-level margins and prices.

TABLE 6
IV STATION-LEVEL ESTIMATES: PRETRENDS

OUTCOME	2SLS		REDUCED FORM	
	Mean Margin (1)	Mean Price (2)	Mean Margin (3)	Mean Price (4)
Adopter $\times$ postadoption	.013*** (.003)	.009*** (.003)		
Adopter $\times$ 1–6 months preadoption	–.000 (.001)	.000 (.001)		
Adopter $\times$ 7–12 months preadoption	.001 (.001)	.001 (.001)		
Adopter $\times$ 13+ months preadoption	.000 (.001)	.002 (.002)		
Share brand adopters $\times$ postadoption			.007*** (.002)	.005*** (.002)
Share brand adopters $\times$ 1–6 months preadoption			–.002 (.003)	–.000 (.004)
Share brand adopters $\times$ 7–12 months preadoption			.005 (.004)	.006 (.005)
Share brand adopters $\times$ 13+ months preadoption			.003 (.010)	.017 (.012)
Station fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	290,585	290,585	290,585	290,585

NOTE.—Sample is gas station/month observations with a history of more than 12 months in the data. Margins are computed above wholesale gasoline prices at a regional terminal nearest to station i . Mean margin/price is the monthly average pump price for station i in month t . “Adopter $\times$ postadoption” is a dummy equal to 1 in month t for stations labeled as adopters after they adopted. “Adopter $\times$ Xmonths preadoption” is a dummy equal to 1 for stations that we labeled as adopters in the X months prior to their adoption. “Share brand adopters” interactions are the excluded instrument used in the 2SLS regression in cols. 1 and 2. They are equal to the share of stations that belong to the brand of station i that adopted by period t , interacted with dummies reflecting whether the station has adopted AP or whether it is going to adopt AP in the future. Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the number of stations belonging to station i 's brand in month t for the number of competitors in the market and for the number of adopting competitors in the market. Weather controls include the mean and standard deviation of monthly temperature and precipitation near station i in month t . Standard errors (in parentheses) are clustered at the market level.

*** $p < .01$.

B. Impact of Adoption on Competition

Section V.A presented causal estimates of the effects of AP on station-level prices and margins. AP can increase margins and prices through a reduction in competition and increased market power, but there can

be other reasons for such changes. An algorithm could better understand underlying fluctuations in wholesale prices or identify how demand elasticity changes over time and adjust prices accordingly. In this section, we describe how we isolate the effects of adoption on competition. We start by splitting the sample according to whether the station is a monopolist (e.g., without any nearby competitors). If adoption affects competition, the impact may be stronger for nonmonopolists than for monopolists. Since there is a lack of clear theoretical predictions on how adoption should affect average prices in monopoly versus nonmonopoly markets if its only function is to improve a seller's ability to tailor prices to time-varying demand or cost conditions,⁶² in a second step we evaluate how adoption affects strategic interaction between stations holding market structure roughly constant by focusing on small oligopoly markets (duopoly and triopoly) and testing whether the adoption of only a subset or of all competitors triggers changes in outcomes.

1. Impact of Adoption on Monopolist and Nonmonopolist Stations

To test whether any observed changes in prices and/or margins come from a reduction in competition and increased market power or from a better understanding of underlying fluctuations in wholesale prices and consumers' demand elasticity, we look separately at stations that are monopolists and stations that are not.⁶³ If adoption does not change competition but benefits station operations in other ways, we might expect to see effects for monopolist adopters. If adoption also affects competition, we should expect to see additional nonzero effects for nonmonopolist adopters on top of the effects for monopolist adopters. If adoption affects only competition, we should expect to see zero effects for monopolist stations and nonzero effects for nonmonopolists.

Results of our IV regression for the two subsamples are presented in table 7. We find that nonmonopolist stations are driving the margin increase, with mean margins increasing for nonmonopolist adopters by 1.2 cents postadoption (15%) compared with a small and non-statistically significant change for monopolist adopters. Price effects are similar: mean monthly

⁶² Monopolists can typically achieve higher profits by price discriminating. Extending these findings to the case of oligopoly is not straightforward since, in addition to market-level elasticities, firm-level elasticities must also be taken into account (see Holmes 1989). As a result, firms might even be worse off under price discrimination than under uniform pricing (see Thisse and Vives 1988; Corts 1998). More generally, there can be considerable variation in potential effects of adoption in both monopoly and oligopoly, depending on (1) what the algorithm does, (2) the informational environment, and (3) supply and demand fundamentals (market- vs. firm-level elasticities).

⁶³ Markets are defined according to a clustering algorithm based on driving time between stations (app. C). As a robustness check, we use an alternative definition based on zip codes. See app. sec. G.2.

TABLE 7
IV STATION-LEVEL ESTIMATES BY MARKET STRUCTURE

	MONOPOLIST STATIONS		NONMONOPOLIST STATIONS	
	Mean Margin (1)	Mean Price (2)	Mean Margin (3)	Mean Price (4)
Adopter × postadoption	.004 (.012)	−.004 (.009)	.012*** (.002)	.013*** (.002)
Nonadopter mean outcome	.0850	1.363	.0825	1.361
Station fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	18,556	18,556	429,181	429,181

NOTE.—Sample includes gas station/month observations from January 2016 to December 2018, split up into two subsamples: one subsample includes only stations that have no competitors in their market, and the other subsample includes only stations that have one or more competitors in their market. Margins are computed above wholesale gasoline prices at a regional terminal nearest to station i . Mean margin/price is the monthly average pump price for station i in month t . “Adopter” is a dummy equal to 1 in month t if the gas station ever experienced a structural break in at least two of four relevant measures, and “postadoption” is a dummy equal to 1 for adopter stations after we label them as adopters. “Share brand adopters” is the excluded instrument used in the 2SLS regression. It is equal to the share of stations that belong to the brand of station i that adopted by period t . Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the number of stations belonging to station i ’s brand in month t for the number of competitors in the market and for the number of adopting competitors in the market. Weather controls include the mean and standard deviation of monthly temperature and precipitation near station i in month t . Standard errors (in parentheses) are clustered at the market level.

*** $p < .01$.

prices increase by 1.3 cpl for adopting nonmonopolist stations but not at all for monopolist stations.⁶⁴

The null estimated effects of adoption for monopolist stations naturally lead to questions about why they would have adopted in the first place. Of course, the null effect only shows that there is no change on the mean. There may well be substantial changes in prices at different points during the day, reflecting a monopolist station’s ability to better price discriminate (as in Dubé and Misra 2023). These changes could average out to a daily null effect. We test for this by using an alternative set of outcomes: mean monthly prices at different points during the day. We calculate the price of each station at 9 a.m., 12 p.m., 5 p.m., and 7 p.m. at each nonholiday weekday and then average out across a month. As mentioned above, gas prices in Germany follow a decreasing pattern throughout the day, with high prices in the morning that gradually fall until the evening. A comprehensive discussion of price cycles in the German gasoline

⁶⁴ In app. sec. G.7, we estimate a pooled version of this regression and results are unchanged.

retail market and the effects of algorithms on these cycles requires formal modeling of both prealgorithmic and post-AP behavior and so is outside the scope of this paper. Nonetheless, our results suggest that, on average, nonmonopolist AP adopters increase their prices during the day such that the price pattern becomes flatter and average daily prices increase.⁶⁵ Monopolist AP adopters show a different pattern, likely reflecting an improved ability to temporally price discriminate and improve overall profits.

IV regression results at different times of the day are presented in table 8. As in table 7, the sample is split between monopolist and nonmonopolist stations (aggregate results are presented in table E4), and, consistent with the findings for mean daily prices, estimates show substantial differences in the effects of AP adoption between monopolists and nonmonopolists. For nonmonopolist adopters, prices do not change in the morning (relative to nonadopters) but then increase progressively throughout the day, with the highest price increase at 5 p.m., generating the flatter pattern we just mentioned. For monopolist adopters, prices fall on average at 9 a.m. relative to nonadopters and increase on average at 5 p.m.. The monopolist results hint at the potential welfare-improving effects of AP adoption through better price targeting across demand conditions. AP software may learn that morning prices are too high and that reducing them will increase monopolist station profits (in addition to consumer welfare). Although human- or rule-based pricing allowed for multiple price changes throughout the day, price adjustments were likely more costly than with AI-powered algorithms, permitting an additional price change (Garcia, Tolvanen, and Wagner 2021). AI-powered algorithms may also help monopolists price discriminate better, which is again profit increasing though not necessarily consumer welfare decreasing. Unfortunately, we do not observe intraday regional wholesale prices and so cannot compute margins.

For nonmonopolist adopters, the potential strategic effects arising from AP could work against the beneficial effects and against targeted price discrimination. Adopting stations may not want to adjust prices to better target consumers if they know competing algorithms are responsive to price changes. This is consistent with our findings that prices are more stable during the day for AP-adopting nonmonopolists. Our findings may also suggest that the value of deviation and punishment are changing throughout the day. The benefits to adopters are stronger later in the day, roughly around the time of the evening commute, possibly implying that without AP, tacit collusion was harder to sustain at this time (although the benefits of adoption are higher even for monopoly stations at 5 p.m.).

Our timing results can be related to the theory literature on temporal price discrimination. Conlisk, Gerstner, and Sobel (1984) explain how a

⁶⁵ It is possible that tacit collusion was more difficult to sustain later in the day when demand was elevated, and so this is when the benefits from adoption are greatest.

TABLE 8
IV STATION-LEVEL ESTIMATES BY MARKET STRUCTURE: TIME-SPECIFIC PRICES

OUTCOME	MEAN PRICE			
	9 a.m. (1)	12 p.m. (2)	5 p.m. (3)	7 p.m. (4)
Monopolist stations:				
Adopter × postadoption	−.029** (.013)	.019 (.014)	.041** (.017)	.012 (.010)
Observations	18,554	18,556	18,556	18,556
Nonadopter mean outcome	1.382	1.358	1.347	1.344
Nonmonopolist stations:				
Adopter × postadoption	−.002 (.003)	.033*** (.003)	.050*** (.004)	.024*** (.003)
Observations	429,094	429,160	429,181	429,181
Nonadopter mean outcome	1.381	1.356	1.345	1.341
Station fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes

NOTE.—Sample includes gas station/month observations from January 2016 to December 2018, split up into two subsamples: one subsample includes only stations that have no competitors in their market, and the other subsample includes only stations that have one or more competitors in their market. Mean price is the monthly average pump price for station i in month t at a particular time. “Adopter × postadoption” is a dummy equal to 1 in month t if the gas station experienced a structural break in at least two of four relevant measures in any previous month $\{1, \dots, t - 1\}$. “Share brand adopters” is the excluded instrument used in the 2SLS regression. It is equal to the share of stations that belong to the brand of station i that adopted by period t . Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the number of stations belonging to station i ’s brand in month t for the number of competitors in the market and for the number of adopting competitors in the market. Weather controls include the mean and standard deviation of monthly temperature and precipitation near station i in month t . Standard errors (in parentheses) are clustered at the market level.

** $p < .05$.

*** $p < .01$.

monopolist can periodically lower its price to discriminate between low- and high-valuation shoppers, the former being more inclined to wait for lower prices than the latter. The monopolist will charge high prices most of the time, selling only to high-valuation shoppers, and will periodically lower its price to attract low-valuation consumers. Sobel’s (1984) model extends Conlisk, Gerstner, and Sobel (1984) to allow for multiple sellers that compete for low-valuation shoppers and finds similar results. Pesendorfer (2002) and Chevalier and Kashyap (2018) develop related models. The pricing patterns in the German retail gasoline market are consistent with these predictions. Consumers with different driving habits—and therefore different inventory costs—can be thought of as having different valuations and more or less incentive to time the price fluctuations. Our

monopoly results imply that AP technologies may make it easier for firms to better coordinate their sales along the daily path.

2. Impact of Adoption on Duopoly/Triopoly Markets

In a more direct test of theoretical predictions about the effects of AP adoption on competition, we compare outcomes between adopting and nonadopting oligopoly markets.⁶⁶ We focus on duopoly and triopoly markets, since most theoretical analysis is done for cases with few firms (i.e., Miklós-Thal and Tucker 2019; Calvano et al. 2020). As with our station-level estimates, we choose to use an IV specification in order to avoid endogeneity concerns. The second-stage regression for market m in month t is as follows:

$$y_{mt} = \alpha_m + \alpha_t + \beta_1 \text{Not All Stations Adopted}_{mt} + \beta_2 \text{All Stations Adopted}_{mt} + \gamma X_{mt} + \epsilon_{mt}, \quad (4)$$

where y_{mt} is the outcome variable for market m at time t and α_m and α_t are market and time fixed effects, respectively. The dummy Not All Stations Adopted is a variable equal to 1 if at least one—but not all—stations in a market are labeled as an adopter at time t . It is equal to zero if all stations are labeled as adopters or if no stations are labeled as adopters.⁶⁷ The variable All Stations Adopted is equal to 1 in market m in month t if all stations in this market are adopters.⁶⁸ The two key coefficients in this regression are β_1 and β_2 . β_1 captures the effects of AP adoption by some of the firms in a duopoly/triopoly market, and β_2 captures the effects of market-wide AP adoption.⁶⁹

⁶⁶ This analysis is done using our main market definition (i.e., clusters). As a robustness check, we use an alternative market definition based on zip codes. See sec. VII and app. sec. G.2 for additional discussion.

⁶⁷ In duopoly markets, this variable can be expressed as $(\text{Adopter} \times \text{Postadoption})_{1mt}(1 - (\text{Adopter} \times \text{Postadoption})_{2mt}) + (\text{Adopter} \times \text{Postadoption})_{2mt}(1 - (\text{Adopter} \times \text{Postadoption})_{1mt})$, where 1 and 2 are the stations in market m and $\text{Adopter} \times \text{Postadoption}$ is a dummy equal to 1 for adopting stations after adoption. The definition for triopoly markets is similar but with a combination of three stations.

⁶⁸ In duopoly markets, this variable can be expressed as $(\text{Adopter} \times \text{Postadoption})_{1mt}(\text{Adopter} \times \text{Postadoption})_{2mt}$, where 1 and 2 are the stations in market m and $\text{Adopter} \times \text{Postadoption}$ is a dummy equal to 1 for adopting stations after adoption. The definition for triopoly markets is $(\text{Adopter} \times \text{Postadoption})_{1mt}(\text{Adopter} \times \text{Postadoption})_{2mt}(\text{Adopter} \times \text{Postadoption})_{3mt}$.

⁶⁹ This distinction is natural for duopoly markets, but triopoly markets can also be separated into those markets where fewer than 50% of stations adopted and markets where more than 50% of stations adopted (e.g., two stations are adopters and one station is not an adopter). We focus on market-wide adoption for two reasons: first, it allows us to aggregate effects across duopoly and triopoly markets. Second, and more importantly, there is substantial evidence that it is harder to sustain supracompetitive prices in markets with asymmetric firms (e.g., with mavericks).

Since there are two endogenous variables, we have two first-stage regressions. Following the logic of our main station-level instruments, we construct two time-varying market-level IVs using brand-level adoption decisions.⁷⁰ In a duopoly market, the two instruments are functions of the brand-level adoption decisions for the brands of stations in market m :

$$\begin{aligned} IV_{mt}^1 &= \text{Share Brand Adopters}_{1mt}(1 - \text{Share Brand Adopters}_{2mt}) \\ &+ \text{Share Brand Adopters}_{2mt}(1 - \text{Share Brand Adopters}_{1mt}), \quad (5) \end{aligned}$$

$$IV_{mt}^2 = \text{Share Brand Adopters}_{1mt} \text{Share Brand Adopters}_{2mt},$$

where $\text{Share Brand Adopters}_{1mt}$ is the share of other stations belonging to market m station 1's brand identified as AP adopters in month t . $\text{Share Brand Adopters}_{2mt}$ is similarly defined.⁷¹

The first-stage regressions are as follows:

$$\begin{aligned} \text{Not All Stations Adopted}_{mt} &= \alpha_m^{1st,1} + \alpha_t^{1st,1} + \pi_1^{1st,1} IV_{mt}^1 + \pi_2^{1st,1} IV_{mt}^2 \\ &+ \kappa^{1st,1} X_{mt} + \mu_{mt}, \\ \text{All Stations Adopted}_{mt} &= \alpha_m^{1st,2} + \alpha_t^{1st,2} + \pi_1^{1st,2} IV_{mt}^1 + \pi_2^{1st,2} IV_{mt}^2 \\ &+ \kappa^{1st,2} X_{mt} + \mu_{mt}. \end{aligned} \quad (6)$$

In each, we include market and time fixed effects and all time-varying controls.

Table 9 presents estimates of equation (4) using the instruments defined in equation (5) and market-level margins and prices as the outcome variables of interest.⁷² The 2SLS estimates are in columns 1 and 2, first-stage estimates are in columns 3 and 4, and reduced-form estimates of regressing the instruments directly on the outcome of interest are in columns 5 and 6. As was the case with the station-level instruments, the partial correlation between market-level instruments and the endogenous variables is strong.

Our estimates suggest that AP adoption by only some stations in duopoly and triopoly markets does not affect average market-level margins or prices relative to similar market where no stations adopted. However, market-wide AP adoption does affect market-level margins and prices.

⁷⁰ As a robustness check for station-level estimates, we propose an alternative instrument: the availability of broadband internet in the local area around a gas station. This instrument would work for market-level data only if the duopolists/triopolists are in the same market but also have different broadband access/quality conditions. Our broadband access data are calculated at a coarse geographical level (NUTS2), so we are unable to use these instruments for market-level data. See additional discussion in app. sec. G.4.

⁷¹ In a triopoly market, if we label $\text{Share Brand Adopters}_{imt}$ as B_{imt} , then $IV_{mt}^2 = B_{1mt}B_{2mt}B_{3mt}$, and $IV_{mt}^1 = B_{1mt}(1 - B_{2mt})(1 - B_{3mt}) + B_{2mt}(1 - B_{1mt})(1 - B_{3mt}) + B_{3mt}(1 - B_{1mt})(1 - B_{2mt}) + B_{1mt}B_{2mt}(1 - B_{3mt}) + B_{1mt}B_{3mt}(1 - B_{2mt}) + B_{2mt}B_{3mt}(1 - B_{1mt})$.

⁷² See app. sec. G.2 for additional discussion of alternative market definitions.

TABLE 9
IV DUOPOLY AND TRIOPOLY MARKET ESTIMATES

OUTCOME	2SLS		FIRST STAGE		REDUCED FORM	
	Mean Market Margin	Mean Market Price	Not All Stations Adopted	All Stations Adopted	Mean Market Margin	Mean Market Price
	(1)	(2)	(3)	(4)	(5)	(6)
Not all stations adopted	−.002 (.005)	.002 (.005)				
All stations adopted	.031* (.018)	.061** (.024)				
IV ¹			.632*** (.088)	−.017 (.025)	−.002 (.003)	.001 (.003)
IV ²			−1.818*** (.476)	1.230*** (.309)	.041** (.017)	.070*** (.019)
Zero adopter mean outcome	.0857	1.355			.0857	1.355
Market fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	49,431	49,431	49,431	49,431	49,431	49,431

NOTE.—The sample includes duopoly and triopoly market/month observations from January 2016 to December 2018. Outcome variable mean market margin is the monthly average of mean market daily differences of pump prices for stations in market m in month t from wholesale price. Outcome variable mean market price is the monthly average of mean market daily pump prices for stations in market m in month t . “Not all stations adopted” is a dummy equal to 1 in month t if at least one station but not all stations in the market experienced a structural break in at least two of four relevant measures in any previous month $\{1, \dots, t - 1\}$. “All stations adopted” is a dummy equal to 1 in month t if all stations in the market experienced a structural break in at least two of four relevant measures in any previous month $\{1, \dots, t - 1\}$. Instruments for adoption are functions of the share of brand adopters of the stations in the market. Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the sizes of the brands of the stations at month t . Standard errors (in parentheses) are clustered at the market level.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

Mean market-level margins increase by 3.1 cpl after market-wide AP adoption. This is a substantial increase of 36% relative to the baseline. Similar effects are observed for market-level prices after market-wide adoption, with mean market prices increasing by 6 cpl.

One explanation for why we do not observe a change in mean market-level margins after incomplete adoption is that the adopter’s margins increase and the nonadopter’s margins fall, canceling out on average. We test this hypothesis by looking at nonadopter stations and comparing

margins and prices before and after their rivals adopt (as before, we instrument for rivals' adoption with the rivals' brand adoption shares). Results are in table E3. There are no statistically significant changes in margins or prices following rivals' AP adoption, ruling out this explanation.

These results serve as a direct test of theoretical hypotheses about the effects of AP adoption on market outcomes (Miklós-Thal and Tucker 2019; Calvano et al. 2020).⁷³ We cannot be sure what type of algorithms stations are using and whether they fully delegate to them pricing decisions. Nonetheless, lack of margin changes from partial/asymmetric adoption and substantial increases in margins and prices after complete adoption indicate that algorithms facilitate tacit collusion. The magnitude of margin increases in duopoly and triopoly markets is consistent with previous findings on coordination in retail gasoline markets (Clark and Houde 2013, 2014; Byrne and De Roos 2019).

VI. Mechanism

In this section, we use data from duopoly and triopoly markets to provide evidence of the mechanism through which algorithmic competition increases prices and margins. We first examine the time it takes for prices to converge to higher, possibly collusive levels following adoption. Updating algorithms operating in fluctuating markets should experience a relatively long adjustment period, as they "learn" and explore the state space, such that convergence to stable strategies can take as long as several years. Asker, Fershtman, and Pakes (2021) show that their less sophisticated asynchronous algorithm converges to something close to the monopoly price but takes considerable time to do so. Calvano et al. (2020) show that convergence to tacitly collusive "punishment" strategies takes time. Alternative explanations of supracompetitive pricing by algorithms do not imply similar temporal patterns.⁷⁴

⁷³ There is also a possibility that multiple stations in a market turn over their pricing decisions to a common algorithmic software provider. Algorithms in this case serve as the hubs of a hub-and-spoke cartel (Garrod, Harrington, and Olczak 2021). If multiple stations in a market turn over their pricing decisions to a common algorithmic software provider, our results are in line with the findings of Decarolis and Rovigatti (2021).

⁷⁴ There are at least two alternative explanations for why algorithms could reach margins above competitive levels. First, they could fail to learn to compete effectively (Cooper, Homem-de-Mello, and Kleywegt 2015; Hansen, Misra, and Pai 2021). Algorithms may not fully incorporate rivals' prices or best respond to them. In this case though, if margins were high, they would remain so initially and then might decrease over time as the algorithms learned to compete. Second, according to Brown and MacKay (2023), adoption of AP software changes the game firms play from a simultaneous Bertrand pricing game to a stage game, thereby increasing prices. We test a key prediction from their model: the bigger the asymmetry in pricing technology, the higher market prices and margins should be. We observe a large number of duopoly markets that feature asymmetric adoption of AP technology. Table E3 shows results from a regression of a nonadopting stations' margins on a dummy variable for whether its rival

TABLE 10
IV DUOPOLY AND TRIOPOLY ADDITIONAL PRICE EFFECTS

OUTCOME	MEAN MARKET WHOLESALE MARGIN (1)	MEAN MARKET PRICE (2)	PROBABILITY OF RESPONSE TO	
			Price Decrease (3)	Price Increase (4)
Months since at least one station adopted:				
0–5	–.001 (.001)	–.000 (.002)	.015 (.021)	–.009 (.019)
6–11	–.002 (.001)	.000 (.002)	.012 (.034)	–.008 (.024)
12+	–.001 (.002)	.002 (.003)	.021 (.078)	–.001 (.052)
Months since all stations adopted:				
0–5	.010* (.005)	.015*** (.005)	.103*** (.022)	.008 (.044)
6–11	.013* (.007)	.022*** (.006)	.102*** (.022)	–.004 (.048)
12+	.045** (.021)	.080*** (.018)	.350*** (.072)	–.046 (.166)
Zero adopter mean outcome	.0857	1.355	.109	.136
Market fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Annual regional demographics	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes
Observations	49,431	49,431	17,337	16,644

NOTE.—The sample includes duopoly and triopoly market/month observations from January 2016 to December 2018. “Months since at least one station adopted” is a dummy equal to 1 in month t if at least one but not all stations in the market has become an adopter in the previous X months and zero otherwise. “Months since all stations adopted” is a dummy equal to 1 in month t if all stations in the market become adopters in the previous X months and zero otherwise. Instruments for both “Months since at least one station adopted” and “Months since all stations adopted” include measures of the share of brand adopters of the stations interacted with timing dummies. Regional demographics include GDP, total population, population density, share of population employed, and median age at the NUTS3/year level. We also control for the sizes of the brands of the stations at month t . Standard errors (in parentheses) are clustered at the market level.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

We provide evidence in favor of this slow convergence to higher margins by examining the timing of adoption effects. Columns 1 and 2 in table 10 show estimates of time-specific effects of incomplete and

has adopted AP technology (instrumented by the rival brand’s adoption share). We find no statistically significant changes in margins following a rival’s adoption. Although the Brown and MacKay (2023) model appears to fit well certain settings (such as cold medicine markets), in our context it does not seem to apply.

complete adoption on mean market margins and prices in a regression that includes the controls from table 9 and market and time fixed effects. The time-specific adoption variables are instrumented by time-specific versions of IV_{mt}^1 and IV_{mt}^2 from equation (5). We bin the timing effects into three periods: the first 6 months after adoption, the second 6 months after adoption, and a year or longer after adoption. We use these bins since there is only a small number of markets we observe for a very long period of time after adoption.⁷⁵

Consistent with simulation results in Calvano et al. (2020) and Asker, Fershtman, and Pakes (2021), we find that for roughly the first year after market-wide AP adoption, there are no statistically significant changes in mean market margins at the 95% confidence level.⁷⁶ The magnitude of estimated coefficients for this time period is also quite small relative to our estimates in table 9, which arise only a year after both stations adopt. For prices, we find similar results. Market-wide prices do increase in the first year after market-wide adoption, but once again the mean effects we estimate in table 9 appear more than a year after market-wide adoption. We find no similar changes in prices or in margins following incomplete adoption. These results are similar to previous findings on transitions to collusive strategies in other markets. For instance, Byrne and de Roos (2019) document a 3-year transition toward coordinated prices in the Australian retail gasoline market.⁷⁷ It is possible that the lagged effects might arise because algorithms take time to learn how to predict and respond to demand and cost shocks. However, unlike the competitive response, these effects are already visible in the first 6 months after adoption. The monopoly effects at 9 a.m. that we document in column 1 of table 8 arise immediately, although they do become a bit stronger over time.

Next, we provide additional suggestive evidence of how algorithmic competition operates differently from nonalgorithmic competition.⁷⁸

⁷⁵ More generally, we have a relatively small number of markets with either partial or complete adoption, which restricts the heterogeneity in effects we can look for in the data.

⁷⁶ Figure 10 in Calvano et al. (2020) shows that profit margins for algorithms do not substantially change for over 500,000 simulation periods. Under the assumption that a simulation period lasts for a few minutes, Calvano et al. (2020) suggest that this would correspond to at least a year. This transition speed is also similar to previous evaluations of algorithmic learning in other settings. For hiring algorithms, Li, Raymond, and Bergman (2021) find that various algorithms require approximately a year to converge to new stable strategies after perturbations in the underlying data.

⁷⁷ Igami and Sugaya (2022) show that 1990s vitamin cartels took several years to increase their prices and margins, while Clark, Horstmann, and Houde (2024) find a lengthy adjustment period to high prices for a Canadian bread cartel. However, these both involve explicit collusion.

⁷⁸ There are no clear conduct measures that can be identified in a reduced form without an underlying model. In our context, developing such a model is not straightforward, since it would require making assumptions about how stations were competing prior to AP introduction and about how algorithms operate. Many algorithms—including Q-learning, as in

We empirically evaluate changes in pricing behavior and the timing of these changes coming directly from duopoly algorithms competing against one another. We focus on pricing patterns that generally characterize AI-powered AP behavior. We know that the algorithms are better than human- or rule-based algorithms at conditioning their behavior on the state of the market and on competitor actions. We test for whether the conditioning behavior evolves differently in markets with and without full AP adoption and whether there is heterogeneity in responsiveness to the direction of competitor price changes. Our two key variables are (1) the market-level probability that if one station reduces its price, another station also reduces its price within 5 minutes; and (2) the market-level probability that if one station increases its price, another station also increases its price within 5 minutes.⁷⁹

Columns 3 and 4 show estimates for the two probabilities. We find that after market-wide adoption, there is an immediate increase in the probability of responding to a rival station's price decrease within 5 minutes. We also find that this propensity is increasing over time, again suggesting that there is gradual learning of new strategies by the algorithms. The magnitude of increased responsiveness is substantial. At the zero adopter baseline, a station has an 11% probability of responding to its rival price decrease with a price decrease of its own within 5 minutes. Twelve or more months after market-wide adoption, the propensity of responding within 5 minutes to a price decrease grows to 50%. The same pattern does not occur in markets where not all stations are AP adopters. Coefficient estimates for markets with incomplete adoption are positive but small and noisy. Notably, this is also not the case for responsiveness to price increases. Column 4 shows no evidence of decreases in stations' propensity to respond to rival price increases after algorithmic adoption.

Together these results are striking and suggest a simple mechanism through which algorithmic competition maintains high prices and margins. Effectively, the algorithms meet any price decrease with an immediate price decrease of their own, teaching each other that undercutting is not profitable since the undercutter will always be followed to the lower price by the other station.

Calvano et al. (2020)—are not designed to play mixed strategies, while others are able to (including humans). There are many possible asymmetric equilibria, and characterizing them without further information is not feasible. We leave this for future research.

⁷⁹ These regressions are related to our adoption marker based on response time to rivals' price changes, but we believe that they are distinct. Here we explicitly allow stations to have both immediate changes in responsiveness (which would identify the initial structural break) and longer-term evolution in responsiveness that identifies changes in competitive strategy. We also separate responsiveness to rival price increases and price decreases.

VII. Robustness

We perform a series of checks to confirm robustness of our results to alternative samples, market definitions, adoption classifications, and instruments. Results with further details are in appendix G. In every case, results on the impact of adoption on margins are robust to the proposed check.

1. Alternative estimation samples (app. sec. G.1): we address possible contamination from Shell's 2015 price matching promotion by (1) dropping observations from markets with Shell stations and (2) dropping all observations from 2016. We also address concerns about entry/exit of stations by using a balanced sample of stations and a balanced sample of stations and markets, dropping any market where the number of stations changes over time.
2. Alternative market definitions (app. sec. G.2): we define a market as a five-digit zip code, a well-defined unit of population in space (rural zip codes are bigger geographically).
3. Alternative adoption definitions (app. sec. G.3): we classify adopters only on the basis of measures that do not rely on the presence of a nearby rival, since this could be important for our comparison of monopoly and nonmonopoly markets. We test a classification that drops responsiveness to crude oil price shocks and another that requires a station to experience a break in the number of price changes. We consider a definition altering the time between structural breaks (in at least two out of four measures within 2 weeks rather than four). We classify a station as an adopter if they break in diesel pricing or in both E5 and diesel.
4. Alternative instruments (app. sec. G.4): we use broadband access in station i 's region as an instrument for adoption. If a station has access to high-speed internet and reliable signals, it should be more likely to adopt AP. We measure whether the local area around the gas station has widespread access to high-speed internet in a given year. We also introduce a placebo instrument. Rather than the share of stations of station i 's brand that adopted as an IV, we use the share of stations of another brand (i.e., the brand of some station k in the market of station i). We expect there to be no correlation between the propensity of station i to adopt and average adoption by other brands, since they do not directly affect station i 's costs.
5. Alternative fuel types (app. sec. G.5): we replicate the analysis, including the definition of adoption, for E10 and Diesel fuels.
6. Alternative specifications (app. sec. G.7): we pool together data for monopolist and nonmonopolist stations when estimating the impact of adoption on prices and margins.

VIII. Policy Discussion and Conclusions

Our findings suggest that regulators should be concerned about mass adoption of AP software. Reports released by antitrust authorities and economic organizations agree that explicit algorithmic collusion would not require change to existing competition law but would affect how authorities monitor and investigate collusive practices. Increased tacit collusion through algorithms could change the legal status of such forms of collusion. Currently, tacit collusion is difficult to prove and prosecute, as it does not rely on explicit communication. The UK Digital Competition Expert Panel (2019, 110) states that with “further evidence . . . of pricing algorithms tacitly co-ordinating of their own accord, a change in the legal approach may become necessary.”

While our evidence is particular to German retail gasoline markets, similar AP software is being adopted in gasoline markets around the world. Our results suggest that authorities should undertake a census of retail gasoline pricing software to understand the structure of the AP software market and the extent of adoption. Such a census can help separate whether the main effect of AP software on competition comes from multiple stations in a market adopting the same or different algorithms. We do not directly observe which algorithm competitors adopt, and the two possibilities have different implications for regulators and policy makers.⁸⁰

Our focus is on the retail gasoline market, but custom-made and off-the-shelf AP software is widely available to use for online and offline retailers. Adoption of such algorithms is growing: Brown and MacKay (2023) present evidence of AP by pharmaceutical drug retailers online. Our results suggest that competition authorities should investigate the relationship between AP software adoption and competition in these and other contexts.

Finally, as mentioned in the introduction, our findings suggest that competition authorities may be focusing their time and resources on the wrong things. Rather than pursuing hard-core cartels on an individual basis, it might be more effective to concentrate on collusion-facilitating devices that do not even require a conspiracy, such as AP and communication via earnings calls (see Aryal, Ciliberto, and Leyden 2022). In a platform setting, Johnson, Rhodes, and Wildenbeest (2023) propose simple market design features that can disrupt algorithmic price-increasing strategies, and such features may have wider applicability in other markets.

⁸⁰ If multiple stations in a market assign pricing decisions to a common algorithmic software provider, our results are in line with the findings of Decarolis and Rovigatti (2021). Algorithms serve as the hubs of a hub-and-spoke cartel (Garrod, Harrington, and Olczak 2021; Clark, Horstmann, and Houde 2024).

Data Availability

Code, publicly available data, and information about obtaining the proprietary data required for replicating the tables and figures in this article can be found in Ershov et al. (2024) in the Harvard Dataverse, <https://doi.org/10.7910/DVN/X4MSWW>.

Appendix A

Structural Break Test Results

Number of price changes.—For each station, we construct a variable measuring the number of times it changes its price for each date in our sample period. For structural break testing, we aggregate this variable to the weekly level.⁸¹ We find that 12,919 stations experience a significant structural break in the number of price changes at the 5% confidence level. Out of the stations that experience significant breaks, almost 50% of the best-candidate breaks occur in the spring and summer of 2017. Figure A1A shows the overall distribution of best-candidate breaks.

Rival response time.—We define a rival for station i as the closest station j that is within a 1-kilometer radius of station i but that belongs to a different brand.⁸² Rival response time for station i is calculated as the number of minutes between the time of a price change by rival j and the subsequent price change by station i . If station i changes its price more than once before station j makes a price change, rival response time is taken as the average of the time gaps between each of station j 's price changes and station i 's subsequent change. When testing for structural breaks in rival response time, we take into account the fact that changes in response time will be mechanically impacted by changes in number of price changes. To identify structural changes separately from this mechanical effect, we control for the number of price changes by both stations. We find that 5,227 experience statistically significant structural breaks. Out of stations with significant breaks (at at least the 5% level), almost 29% have best-candidate breaks in the spring and summer of 2017. Figure A1B shows the overall distribution of best-candidate breaks.

Responsiveness to crude oil price shocks.—We observe an intraday time series for crude oil prices. In each nonholiday weekday, we separate fluctuations in crude oil prices from the moving average. We define a crude oil price shock as large deviations from the moving average. More concretely, they are defined as deviations from the moving average that are above the 90th percentile of all deviations in a given year-month. This helps us account for changing volatility of oil prices over time. We define a response to a crude oil price shock as a price change within 5 minutes of the shock.

The outcome variable in the QLR regressions is the average number of times a station responds to an oil price shock in a week. We control for the average number of price changes a station makes in a week as well as for the number of oil shocks

⁸¹ Any stations that do not have a weekly observation for average number of price changes in every week of 2017 are dropped. See more details in the Replication Package (Ershov et al. 2024).

⁸² This reflects the average distance of stations in the data.

that happen in a week. This helps to control for the fact that oil price volatility is changing throughout our sample. We find that there are 5,747 stations with statistically significant breaks (at the 5% confidence level). Figure A1C shows the overall distribution of best-candidate breaks.

Responsiveness to local weather shocks.—Using data from the DWD, we observe a high-frequency time series of local air temperature around each gas station. We separate fluctuations in temperature from the moving average in each non-holiday weekday. We define a local weather shock as large deviations from the moving average. They are defined as deviations from the moving average that are above the 90th percentile of deviations in a given year-month. We define a response to a local weather shock as a price change within 5 minutes of the shock.

The outcome variable in the QLR regressions is the average number of times a station responds to a local weather shock in a week. We control for the average number of price changes a station makes in a week as well as for the number of weather shocks that happen in a week, meaning that we are allowing for changes in responsiveness conditional on the weather volatility around the station. We find that there are 4,892 stations with statistically significant breaks. Figure A1D shows the overall distribution of best-candidate breaks.

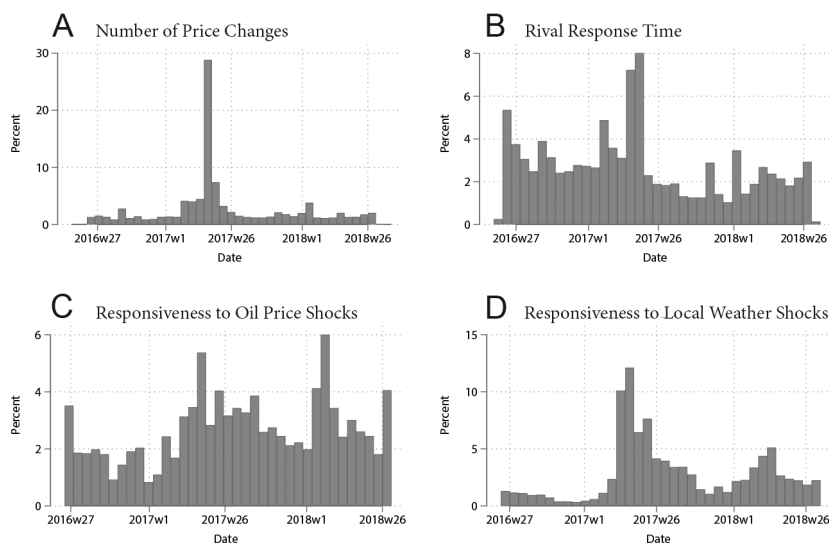


FIG. A1.—Frequency of best-candidate structural breaks. Histograms show the distribution of best-candidate QLR structural break weeks for the number of price changes (12,919 stations included; A), the response time to a rival's price changes (5,227 stations included; B), the number of responses to oil price shocks (conditional on the number of shocks and the number of station price changes; 5,747 stations included; C), and the number of responses to local weather shocks (conditional on the number of shocks and the number of station price changes; 4,892 stations included; D).

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